

A Regime-Aware Model for Loan-Level Probability of Default

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Declaration

I declare that this thesis is my own work towards the Master of Philosophy degree and that, to the best of my knowledge, it contains no material previously published by another person or submitted for another degree, except where due acknowledgement has been made in the text.

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Certification

We certify that this thesis was supervised in accordance with the procedures established by Kwame Nkrumah University of Science and Technology.

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Dedication

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Abstract

This thesis evaluates whether an interpretable macroeconomic regime signal improves twelve-month loan-level mortgage probability-of-default (PD) forecasts. Freddie Mac sample origination and monthly performance records for the 2015–2024 cohorts were linked to public macroeconomic series from Federal Reserve Economic Data. The eligible sample contained 438,713 loans, 15,461,471 loan-months, and 167,056 transitions to at least 90 days past due. A two-state Gaussian hidden Markov model compressed four adverse-oriented macroeconomic factors into a filtered stress probability. That probability entered a regularised logistic model directly and through eight selected interactions with borrower, loan, and payment-history characteristics. Borrower-only and additive macroeconomic logistic models provided the principal comparators. Validation-based Platt scaling supplied the primary calibrated probabilities.

In the original terminal test period, the regime-aware model had a Brier score of 0.006015 and log loss of 0.032945, compared with 0.006055 and 0.033106 for borrower-only. The difference was small, borrower-only remained stronger on validation and monthly default-count calibration, and terminal stress probabilities were nearly saturated at one. A pre-defined causal rolling-origin analysis therefore re-estimated factor transformations, scaling, and HMM parameters only through each training cutoff and evaluated three non-overlapping twelve-month test windows. Across 6,650,858 pooled rolling-test loan-months, borrower-only achieved lower Brier score and log loss, higher ROC-AUC and PR-AUC, and lower monthly count error than regime-aware. Three-month moving-block intervals for paired Brier and log-loss differences included zero. Stress probabilities remained extreme under causal fitting.

The evidence does not establish universal predictive superiority for the regime-aware model. Its advantage was period-dependent: it improved one causal origin and the original terminal split, but the gain did not persist across origins. The model is therefore retained as an interpretable diagnostic of changing borrower sensitivities under fitted macroeconomic conditions, while borrower-only logistic regression is the more stable pooled forecasting model in this study.

Keywords: probability of default; hidden Markov model; mortgage credit risk; calibration; rolling-origin validation; regime awareness.

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List of Abbreviations and Symbols

Term	Meaning
PD	Probability of default
DPD	Days past due
HMM	Hidden Markov model
FRED	Federal Reserve Economic Data
ROC-AUC	Area under the receiver operating characteristic curve
PR-AUC	Area under the precision–recall curve
CPI	Consumer Price Index
FHFA	Federal Housing Finance Agency
NFCI	National Financial Conditions Index
$Y_{i,t}^{(12)}$	Indicator that loan i reaches at least 90 DPD within twelve months after month t
$X_{i,t}$	Borrower, contract, account, and payment-history feature vector
M_t	Vector of transformed observed macroeconomic variables
R_t	Latent macroeconomic HMM state
q_t	Forward-filtered probability that month t belongs to the stress state
$W_{i,t}$	Selected loan-level variables interacted with q_t
$\hat{\pi}_{i,t}$	Calibrated predicted twelve-month PD

Introduction

1.1 Background and Problem

Loan-level probability of default (PD) is the conditional probability that a particular loan enters a defined default state within a specified horizon, given the information available at the prediction date. This study uses a loan-month prediction unit and a twelve-month serious-delinquency outcome. For a loan i observed in month t , the outcome is

$$Y_{i,t}^{(12)} = \mathbb{I}\left\{\max_{h=1,\dots,12} D_{i,t+h} \geq 90\right\}, \quad (1.1)$$

where $D_{i,t+h}$ is days past due and the loan has not already reached 90 days past due at month t . The corresponding PD is

$$\pi_{i,t}^{(12)} = \Pr\left(Y_{i,t}^{(12)} = 1 \mid \mathcal{F}_t\right), \quad (1.2)$$

where $\mathcal{F}_t$ contains information available by month t . The event is therefore the first transition to 90 or more days past due during months $t + 1$ to $t + 12$. This threshold is an operational measure of serious mortgage delinquency and is consistent with regulatory treatment of material arrears as evidence of default (Basel Committee on Banking Supervision, 2004; Board of Governors of the Federal Reserve System, 2006).

Loan-level PD models must estimate probability levels as well as rank loans by risk. Discrimination measures whether higher-risk loans tend to receive higher scores. Calibration

measures agreement between predicted probabilities and observed event frequencies. Both properties matter because a well-ranked score can still overstate or understate the level of risk (Brier, 1950; Gneiting and Raftery, 2007; Van Calster et al., 2019).

Logistic regression is a standard loan-level credit-scoring method. It provides an explicit probability function and coefficients with a direct log-odds interpretation (Hand and Henley, 1997; Thomas, 2000; Costa e Silva et al., 2020). A borrower-only specification, however, assumes that the relationship between borrower characteristics and default remains stable through calendar time. This restriction can be material when common economic conditions change the consequences of leverage, payment burden, interest rates, or recent delinquency.

Observed macroeconomic variables can be added to the logistic equation. Such models connect loan-level risk to the credit cycle and are useful in stress analysis (Bellotti and Crook, 2009, 2013). A purely additive specification still assumes that macroeconomic variables shift default risk without changing borrower sensitivities. It also represents a persistent and multidimensional economic state as a linear combination of contemporaneous indicators. These restrictions may be inadequate when the same borrower characteristic carries different risk under benign and stressed conditions.

1.2 Research Gap and Aim

The literature contains established loan-level scorecards, macro-augmented credit models, and latent-state credit-risk models. Their intersection is narrower. Borrower-only models often impose constant coefficients. Additive macroeconomic models do not allow the effect of borrower information to vary with the economic state. Regime-switching applications frequently address rating migrations, default intensities, or portfolio-level quantities rather than an interpretable loan-month PD (Bangia et al., 2002; Yu et al., 2018). This leaves an empirical question about whether a filtered macroeconomic regime probability can improve a transparent loan-level model through selected borrower–regime interactions.

The study aimed to develop and evaluate a regime-aware model for twelve-month mortgage PD. A two-state Gaussian hidden Markov model (HMM) reduced a set of public macroeconomic indicators to a filtered probability of stress. That probability entered a regularised logistic model directly and through selected interactions with borrower, loan, and payment-history information.

1.3 Objectives and Research Questions

The completed study had three objectives:

1. It estimated macroeconomic regime probabilities using a two-state HMM.
2. It developed a regime-aware loan-level PD model that combined borrower information with the estimated regime probabilities.
3. It evaluated the regime-aware model against borrower-only and macro-augmented logistic baselines.

The corresponding research questions are:

1. What macroeconomic stress probabilities did the two-state HMM assign over the study period?
2. How did the inclusion of the stress probability and borrower–regime interactions alter the loan-level PD specification?
3. How did the regime-aware model compare with borrower-only and macro-augmented logistic baselines on calibration and discrimination?

1.4 Contribution and Scope

The methodological contribution is an interpretable link between a two-state macroeconomic HMM and a regularised loan-level logistic model. The regime probability is a continuous in-

put. Selected interactions allow specified borrower effects to vary with that probability. The empirical contribution is an application to Freddie Mac mortgage origination and monthly performance data matched to public macroeconomic series. The evaluative contribution is a chronological comparison based on the Brier score, log loss, calibration diagnostics, ROC-AUC, PR-AUC, and monthly default-count error.

These contributions are limited in scope. The study does not claim that regime switching is new in credit risk. It tests a specific interpretable specification on one United States mortgage dataset. The original terminal comparison estimated HMM parameters and factor transformations over the full study window, although the reported state probabilities were generated by forward filtering. A final robustness pass replaced that feature path with three causal rolling origins in which transformations and HMM parameters were fitted only through each training cutoff. Stress probabilities nevertheless remained extreme, and model rankings changed across origins. These features restrict cross-regime and external interpretation.

Chapter 2 reviews the evidence directly relevant to the model. Chapter 3 defines the data, HMM, logistic specifications, terminal and rolling validation designs, and evaluation measures. Chapter 4 reports the retained empirical results and answers the research questions. Chapter 5 interprets the joint evidence and its limitations. Chapter 6 states the conclusions and bounded recommendations. The appendices document variables, specifications, full rolling diagnostics, and reproducibility information.

Literature Review

Probability of default is conditional on an event definition, a horizon, a risk set, and an information set. These elements determine the estimand. In a loan-month model, the relevant quantity is $\pi_{i,t}^{(12)} = \Pr(Y_{i,t}^{(12)} = 1 \mid \mathcal{F}_t)$, where the outcome records whether loan i first reaches serious delinquency during the following twelve months. A change in the delinquency threshold, prediction horizon, treatment of terminated loans, or information available at month t changes the probability being estimated. PD should therefore be interpreted as a forecast for an eligible loan at a defined date rather than as an unconditional borrower attribute.

This conditional interpretation separates PD estimation from the retrospective description of defaulted loans. A retrospective analysis asks which characteristics are common among realised defaults. A PD model asks how those characteristics alter the probability of a future event among loans that remain eligible to default. The risk-set restriction is part of that question. Including observations that are already seriously delinquent changes the task from transition prediction to state classification. Treating loans with incomplete follow-up as non-events changes the observed event frequency and can bias calibration. Loan-level PD estimation therefore requires the event construction and the eligible population to be stated together.

The horizon also affects the meaning of the probability. A one-month delinquency probability and a twelve-month serious-delinquency probability are not interchangeable. The longer horizon aggregates the chance of an event over several future months and generally

produces a higher cumulative event rate. It also increases the importance of adequate follow-up. The twelve-month horizon used in this study is fixed for every loan-month, which permits a direct comparison between predicted values and realised frequencies. It does not estimate a lifetime default probability or a time-to-event hazard.

Logistic regression remains a central method for loan-level credit scoring. It maps a linear predictor to the unit interval and provides coefficients that describe changes in log-odds. Credit-scoring research has used logistic models for both origination characteristics and behavioural information observed after account opening (Hand and Henley, 1997; Thomas, 2000; Costa e Silva et al., 2020). Origination variables describe initial credit quality, leverage, balance, interest rate, term, and borrower structure. Monthly performance variables describe the subsequent payment path. Recent delinquency, its duration, and evidence of cure can contain information that is not available in the origination record.

The logistic link provides a coherent probability only when the fitted linear predictor is combined with an appropriate estimation and calibration procedure. The coefficient on a continuous variable describes a change in log-odds holding the other included variables constant. This conditional interpretation is useful, but it is sensitive to the set of correlated predictors and their scaling. A coefficient should not be read as a causal effect unless the design supports that claim. In a predictive PD model, its principal role is to make the contribution of a defined feature explicit.

Origination and behavioural variables serve different purposes. Origination characteristics measure the risk profile at the start of the loan. Monthly balance, interest rate, age, remaining term, and delinquency history update that profile as the account evolves. Behavioural variables can dominate short-horizon prediction because they record deterioration that has already begun. Their construction must stop at the prediction month. A rolling count that incorporates month $t + 1$, even indirectly, would contaminate the information set used for a month- t forecast.

Repeated loan-month observations create another distinction from a cross-sectional score-

card. The same loan contributes several observations, and its static characteristics recur while its age, balance, payment record, and common macroeconomic environment change. The resulting sample can be large without containing an equally large number of independent economic periods. Chronological validation is therefore more informative for temporal transport than a random division of loan-month rows. Random allocation would place observations from the same calendar month, and often the same loan, on both sides of the evaluation boundary.

A borrower-only logistic model is static in calendar time unless temporal terms are introduced explicitly. Its coefficients imply that a unit change in a borrower or loan characteristic has the same log-odds effect in all months. This is a strong restriction for mortgage PD. The risk associated with leverage, interest cost, or recent delinquency may differ when labour income, financing conditions, and collateral values deteriorate together. A static scorecard can retain useful cross-sectional discrimination while its probability levels drift as the economic environment changes.

Static coefficients can fail in two related ways. The intercept may no longer represent the prevailing base rate, and the slopes may no longer represent the prevailing relationship between loan characteristics and the outcome. Recalibrating the intercept can address a broad level shift. It cannot correct a change in the relative importance of credit score, leverage, interest cost, or payment history. A model that allows only an intercept adjustment remains static in its borrower sensitivities.

The distinction matters for evaluation. If every predicted probability is too high by a similar amount, the model can preserve ROC-AUC while producing poor Brier score and log loss. If the slopes change, ranking can also deteriorate. A complete assessment should therefore compare probability scores, calibration levels, and discrimination. A single measure cannot identify whether error arises from a base-rate shift, weak separation, or both.

Macroeconomic credit models address part of this problem by adding observed economic variables to the loan-level equation. Unemployment, interest rates, inflation, house prices,

and related variables connect default forecasts to common conditions. Evidence from survival and stress-testing models shows that macroeconomic variables can improve the temporal responsiveness of retail credit models (Bellotti and Crook, 2009, 2013). The usual additive form shifts the log-odds by the same macroeconomic contribution for all loans. It does not imply that a loan characteristic becomes more or less consequential as the economy moves into stress. A large correlated macro vector can also weaken stability and interpretation.

An additive macroeconomic model has the form $\text{logit}(\pi_{i,t}) = \beta_0 + X_{i,t}^\top \beta + M_t^\top \gamma$. The same $M_t^\top \gamma$ is applied to every loan observed in month t . Borrowers can still receive different PDs because their $X_{i,t}$ values differ, but the marginal log-odds effect of each borrower feature remains β_j in every month. The macroeconomic vector changes the common level of risk rather than the cross-sectional sensitivity to borrower information.

This form is useful as a baseline because it tests whether direct macroeconomic measurements add information after loan-level variables are included. It is also demanding to estimate when several indicators describe related economic mechanisms. Policy rates, mortgage rates, Treasury yields, and spreads are correlated. Unemployment, claims, payrolls, income, and sentiment overlap in their description of labour and household conditions. Regularisation can stabilise their coefficients, but the fitted vector can remain difficult to summarise economically.

The timing of macroeconomic variables must match the information set. Levels, changes, spreads, and growth rates use different amounts of historical information. A twelve-month growth rate at month t is available from observations at t and $t - 12$. A monthly change is available from t and $t - 1$. These backward-looking transformations are consistent with a month- t forecast. Revisions to public macroeconomic series create a separate real-time issue when current-vintage data are used to describe earlier months. The present study does not reconstruct historical data vintages, so its macroeconomic inputs should be interpreted as revised series aligned by calendar month.

Hidden Markov models provide a parsimonious representation of persistent latent condi-

tions. The observed macroeconomic vector is assumed to arise from one of a finite number of unobserved states, while the state follows a first-order Markov chain. State persistence distinguishes an HMM from a contemporaneous classification rule. The model can summarise several indicators as a probability distribution over economic states rather than assigning an observed month to a regime without uncertainty (Rabiner, 1989; Hamilton, 1989).

A two-state HMM separates the macroeconomic sequence into two fitted distributions. The state labels are statistical and have no inherent economic meaning. One state becomes Stress only after its fitted emission profile is compared with the orientation of the inputs. This post-estimation orientation avoids assigning meaning to the arbitrary numeric state labels. It also makes the definition reproducible: the stress state is the state with the larger combined mean across variables that have been oriented towards adverse conditions.

The transition matrix represents persistence. Large diagonal probabilities imply that the latent state tends to remain unchanged from one month to the next. This persistence can reduce month-to-month noise in the regime signal. It can also produce probabilities close to zero or one when the fitted state distributions are well separated. Such certainty is not automatically evidence of a superior economic signal. Near-degenerate probabilities may leave too little variation for interaction effects to be distinguished in a later test period.

The Gaussian emission assumption is a modelling restriction. Within each state, the factor vector is represented by a multivariate normal distribution with a state-specific mean and covariance matrix. Full covariance matrices allow the relationships among factors to differ by state. They require more parameters than diagonal covariance matrices and can be sensitive when the number of monthly observations is limited. Reducing the macroeconomic inputs to a small number of interpretable factors limits this parameter burden without claiming that the factors exhaust every dimension of the macroeconomy.

The distinction between filtered and smoothed state probabilities is essential in forecasting. A smoothed probability conditions on the complete observed sequence and therefore uses information after month t . A filtered probability conditions on observations through

month t . The filtered stress probability, denoted $q_t = \Pr(R_t = \text{Stress} \mid Z_{1:t}; \hat{\Theta})$, is the appropriate state output for a month- t PD equation. It remains conditional on the method used to estimate $\hat{\Theta}$. If HMM parameters and transformations are fitted over the full sample, forward filtering prevents direct use of future observations in the recursion but does not create a fully real-time estimate.

This distinction produces two separate information questions. The first concerns state inference given fixed parameters. Forward filtering answers that question using $Z_{1:t}$. The second concerns parameter estimation. A fully real-time design would estimate transformations, scaling parameters, and HMM parameters using information available at each forecast origin. Using a full-window estimate answers a narrower question: what filtered state path is implied by parameters fitted to the complete study window? The resulting signal is more defensible than a smoothed path for downstream prediction, but it should not be described as strictly out of sample.

The continuous probability q_t contains more information than the hard indicator $\mathbb{I}(q_t \geq 0.5)$. Values near 0.5 express state uncertainty. Values near zero or one express stronger assignment under the fitted model. A hard threshold discards this gradation and can create an artificial discontinuity. It is still useful for descriptive counts and diagnostic slices. It is unsuitable for a normal-versus-stress comparison when every observation in an evaluation period lies on the same side of the threshold.

Regime-dependent credit models allow default processes to vary across persistent economic states. Existing applications have examined cyclical rating migration and hidden-state default intensity (Bangia et al., 2002; Yu et al., 2018). This literature establishes that latent regimes can enter credit-risk models, but it does not settle whether an HMM improves a loan-level mortgage PD after borrower and payment-history information is included. It also leaves a choice between a hard regime indicator and a continuous state probability. A continuous probability preserves uncertainty and avoids an abrupt change at an arbitrary threshold.

Portfolio-level regime models and loan-level regime-aware models answer different questions. A portfolio model can allow a common transition matrix or default intensity to change with the state. A loan-level model must also represent cross-sectional heterogeneity among loans observed in the same month. The common state alone cannot distinguish two borrowers. Borrower information remains the primary source of cross-sectional variation, while the regime signal supplies a shared time-varying context.

This layered interpretation places the HMM between the macroeconomic data and the PD equation. It is a dimension-reduction and state-inference component rather than a replacement for the credit model. The HMM is estimated from monthly macroeconomic factors. The logistic model is estimated from loan-month outcomes. Their parameters have different meanings and are supported by different effective sample sizes. Millions of loan-months do not eliminate uncertainty in a regime process estimated from slightly more than a decade of monthly observations.

A regime-aware loan-level logistic model can include q_t as a level effect and interact it with selected components of $X_{i,t}$. The log-odds contribution of feature j then becomes $\beta_j + q_t\theta_j$. This structure is more flexible than an additive macroeconomic model and remains interpretable by construction. Interpretation still requires care. The sign and magnitude of a standardised interaction coefficient depend on the included main effects, feature scaling, and correlation among variables. The structure identifies which sensitivities are allowed to vary; it does not by itself establish a causal effect of the macroeconomic state.

The level term and interaction terms serve distinct purposes. The coefficient on q_t shifts the log-odds for every loan after accounting for $X_{i,t}$. Each interaction changes the coefficient on one selected loan-level feature as q_t changes. A stress-level-only model can therefore be compared with the interaction model. If the level-only model improves but the interaction model does not, the evidence favours a common macroeconomic shift. If the interaction model improves while the level-only model does not, the selected changing sensitivities are the more plausible source of the predictive difference.

Interaction selection should remain limited when interpretation is a central aim. Interacting every loan-level variable with every macroeconomic series would create a large model whose coefficients are difficult to separate. A smaller set based on credit quality, leverage, interest cost, and recent payment performance preserves the economic question. Regularisation remains necessary because the main effects and their products with a common q_t can be strongly correlated, especially when q_t spends long periods near an endpoint.

When q_t has little variation in the test period, the regime-aware model can still generate different predictions because the interaction coefficients were estimated using earlier variation and because borrower features vary across loans. Its held-out performance can therefore be measured. What cannot be established is performance across distinct test regimes. A favourable pooled score during an all-stress period is evidence about that terminal period, not evidence of a stable normal-to-stress contrast.

Interpretability is relevant because a higher-capacity specification can improve fit while obscuring the source of a probability change. A regularised logistic model retains an explicit functional form and permits direct inspection of borrower–regime terms. This advantage does not exempt the model from empirical validation. A transparent model can still be unstable, poorly calibrated, or sensitive to the estimation window. Interpretability concerns the form of the model. Predictive validity concerns its behaviour on data not used for estimation.

Regularisation changes the interpretation of fitted coefficients. An L_2 penalty shrinks correlated coefficients towards zero but does not set most of them exactly to zero. The resulting model can be more stable than an unpenalised fit, particularly with macroeconomic and interaction terms. Coefficients remain conditional associations on the standardised scale. Their presence documents the modelled pathway from a predictor to log-odds; their magnitude should not be presented as an unpenalised structural parameter.

Interpretability also depends on reporting the complete specification. A label such as “regime-aware” is insufficient unless the main effects, regime input, and interaction set are defined. The calibration map is part of the final probability model because it changes

probability levels after the logistic fit. A transparent account should therefore distinguish the raw logistic probability, the calibration transformation estimated on validation data, and the final probability used for evaluation.

Calibration and discrimination answer different questions. ROC-AUC measures the ordering of cases and controls, while PR-AUC focuses more directly on positive-class retrieval when defaults are rare (Davis and Goadrich, 2006; Saito and Rehmsmeier, 2015). Neither measure assesses whether predicted probabilities match observed frequencies. The Brier score and log loss are proper scoring rules that assess probability forecasts (Brier, 1950; Gneiting and Raftery, 2007). Calibration curves compare predicted and observed rates across probability groups. A Brier decomposition separates reliability, resolution, and outcome uncertainty, although its numerical values depend on the grouping rule.

The Brier score averages squared probability errors. It is sensitive to calibration and discrimination but is often dominated by the large number of non-events when the outcome is rare. Log loss gives a larger penalty to confident errors because it is logarithmic. The two scores can rank models differently. Agreement between them strengthens a comparison, while disagreement requires a metric-specific interpretation. Neither score should be converted into a claim of economic significance without considering the size of the difference and the dependence among repeated observations.

ROC-AUC is invariant to monotonic recalibration. A model can retain the same ROC-AUC after its probabilities are shifted substantially. PR-AUC is also primarily a ranking measure, but its scale depends on event prevalence. Reporting the event rate makes its magnitude interpretable. In a rare-event setting, PR-AUC can distinguish models whose ROC curves appear similar because it places more emphasis on the precision of high-ranked positive predictions.

Calibration has several levels. Mean calibration compares the average predicted PD with the overall event rate. Calibration-in-the-large can be poor even when a curve has acceptable local shape. A calibration slope assesses whether probabilities are too dispersed

or too compressed. Grouped curves show how predicted and observed rates compare across selected bins. These summaries complement one another. A model can have a lower Brier score than a comparator while both models materially overpredict the mean event rate.

The Murphy decomposition expresses a grouped Brier score as reliability minus resolution plus uncertainty. Reliability is smaller when average predictions within bins are close to observed rates. Resolution is larger when observed rates differ across bins. Uncertainty depends only on the overall event rate. Because the decomposition uses bins, it approximates rather than reproduces the ungrouped Brier score. Its components should be used to explain a result, not to replace the primary ungrouped score.

Post-estimation calibration can correct systematic distortion in raw probabilities. Platt scaling estimates an intercept and slope on the logit of the raw probability. Intercept-only recalibration adjusts the level while preserving the slope. Isotonic regression estimates a non-decreasing mapping with greater flexibility. These methods should be fitted without using the test outcomes. Chronological validation is therefore important for both model selection and calibration. It also reveals whether a result is stable across time rather than a consequence of random mixing across economic periods.

Platt scaling can correct both a broad level shift and excessive or insufficient dispersion on the logit scale. Intercept-only recalibration is more restrictive and leaves ranking and spread unchanged. Isotonic regression makes fewer shape assumptions but can adapt closely to the validation sample. Comparing these mappings on the same held-out period provides a sensitivity analysis. Choosing a new primary method after inspecting test outcomes would weaken the separation between validation and evaluation.

Temporal validation carries information beyond a single test score. A model that is best on validation and worse on test may have deteriorated as the population or economy changed. A model that is worse on validation and better on test may be suited to the terminal period without demonstrating stable superiority. Both patterns should be reported. The final interpretation should distinguish a held-out result from a recurring result across evaluation

windows.

Portfolio calibration adds another level of assessment. Summing loan-level PDs within a month gives an expected number of defaults under the model. Comparing that quantity with the observed monthly count tests whether pooled probability performance translates into useful timing and level estimates. A model can improve the average Brier score by reallocating risk across loans while still overstating the total number of defaults. This is not a contradiction. It reflects different aggregations of the same probability forecasts.

The evidence supports a focused empirical comparison. Borrower-only logistic regression provides the loan-level benchmark. An additive macroeconomic logit tests whether observed macro variables shift PD levels. A regime-aware logit tests whether a filtered stress probability and selected borrower–regime interactions improve probability forecasts. The comparison should place calibration and proper scores alongside discrimination. It should also examine monthly predicted-default counts because a model can improve pooled loan-level scores while misstating the timing or level of portfolio risk.

This comparison gives each model a defined role. The borrower-only model establishes the value of cross-sectional and behavioural information without common economic inputs. The macro-augmented model tests a direct additive treatment of observed conditions. The regime-aware model tests a compressed latent-state signal and changing sensitivities. Diagnostic variants can separate the stress level from its interactions and assess whether the full macro vector contributes beyond the regime probability. The empirical conclusion should follow the joint pattern of these comparisons rather than the label attached to any one specification.

Methodology

3.1 Data and Study Population

The study combined Freddie Mac Single-Family Loan-Level Dataset sample files with macroeconomic series from Federal Reserve Economic Data (FRED) (Freddie Mac, 2026; Federal Reserve Bank of St. Louis, 2026). Freddie Mac origination records supplied initial borrower and contract characteristics. Monthly performance records supplied delinquency status, balance, interest rate, loan age, remaining term, and termination information. The available sample cohorts covered 2015–2025. The twelve-month follow-up rule left eligible observations from the 2015–2024 cohorts and a common endpoint of September 2024.

Table 3.1: Data components and roles in the retained specification

Component	Information	Empirical role
Freddie Mac origination records	Credit score, leverage, balance, rate, term, insurance, units, and borrower count	Initial borrower and contract risk characteristics
Freddie Mac monthly performance records	Delinquency status, balance, current rate, loan age, remaining term, and termination	Risk set, twelve-month outcome, and time-varying loan features
FRED macroeconomic series	Prices, rates, labour indicators, income, house prices, financial conditions, and sentiment	Transformed macro baseline and four HMM factors
Filtered HMM output	Monthly probability of the stress state	Direct regime term and borrower–regime interactions

Origination and performance files were linked by the loan sequence identifier. Calendar

dates were normalised to the first day of each month. Delinquency status in the servicing data was converted from months delinquent to days past due by multiplying non-negative numeric values by 30. Loan-level records were ordered by identifier and month before the outcome and payment-history variables were calculated.

The unit of analysis was a loan-month. Let $D_{i,t}$ denote days past due for loan i in month t . A loan-month entered the risk set when the loan was active, $D_{i,t} < 90$, the servicing record extended through $t + 12$, and no terminal servicing event occurred on or before $t + 12$. The eligible set was

$$\mathcal{R} = \{(i, t) : D_{i,t} < 90, t + 12 \leq T_i^{\max}, T_i^{\text{term}} > t + 12 \text{ or } T_i^{\text{term}} \text{ is missing}\}, \quad (3.1)$$

where $T_i^{\max}$ is the last observed month and T_i^{term} is the first terminal month. This rule excluded observations with incomplete forward follow-up instead of treating them as non-events.

The primary outcome was

$$Y_{i,t}^{(12)} = \mathbb{I}\left\{\max_{h=1,\dots,12} D_{i,t+h} \geq 90\right\}, \quad (i, t) \in \mathcal{R}. \quad (3.2)$$

The conditional PD was $\pi_{i,t}^{(12)} = \Pr(Y_{i,t}^{(12)} = 1 \mid \mathcal{F}_t)$. All predictors were measured at origination or constructed from information observed by month t .

3.2 Loan-Level Predictors

The borrower and loan vector $X_{i,t}$ contained 14 origination or current-account variables and 18 payment-history variables. The first group comprised credit score, mortgage-insurance percentage, number of units, original combined loan-to-value ratio, original debt-to-income ratio, original unpaid principal balance, original loan-to-value ratio, original interest rate, original term, number of borrowers, loan age, remaining months to legal maturity, current

unpaid principal balance, and current interest rate.

The payment-history group comprised current delinquency in months; current 30- and 60-day delinquency indicators; recent 30- and 60-day delinquency within three months; ever-30- and ever-60-day indicators; months since 30- and 60-day delinquency; counts of 30- and 60-day delinquent months over three, six, and twelve months; a cure-from-30-days indicator; cumulative 30-day cures; and months since a 30-day cure. These features were calculated within loan and used observations through month t .

Table 3.2: Loan-level predictor groups

Group	Variables
Origination and account	Credit score; mortgage-insurance percentage; number of units; original combined and ordinary loan-to-value ratios; original debt-to-income ratio; original and current unpaid principal balance; original and current interest rate; original term; number of borrowers; loan age; remaining months to legal maturity
Current delinquency	Delinquency months; current 30-day indicator; current 60-day indicator
Recent and cumulative history	Recent 30- and 60-day indicators; ever-30- and ever-60-day indicators; months since 30- and 60-day delinquency
Rolling history	Counts of 30- and 60-day delinquent months over three, six, and twelve months
Cure history	Cure from 30 days past due; cumulative 30-day cures; months since a 30-day cure

The rolling counts were based on cumulative within-loan event counts and their lags. A recent indicator equalled one when the corresponding three-month count was positive. A cure occurred when a loan moved from at least 30 days past due in month $t - 1$ to below 30 days past due in month t . For loans with no prior delinquency or cure, the elapsed-month variable was measured from the first observed loan month plus one. This convention provided a finite history measure without using a later event date.

3.3 Macroeconomic Variables and Factors

The macroeconomic data contained CPI, the Federal Funds rate, unemployment, the FHFA house-price index, the 30-year mortgage rate, the 10-year Treasury rate, the 10-year minus 2-year Treasury spread, initial unemployment claims, non-farm payrolls, real disposable personal income, the National Financial Conditions Index, and consumer sentiment. Weekly and daily observations were averaged to monthly frequency. The retained transformations were twelve-month inflation, the policy rate, unemployment and its monthly change, twelve-month house-price growth, the mortgage rate, the mortgage–Treasury spread, the long rate and its monthly change, the yield curve, initial claims and their monthly change, twelve-month payroll and real-income growth, financial conditions, consumer sentiment, and its monthly change.

The HMM did not receive the 17 transformed series directly. Each series was oriented so that a larger value represented greater stress and standardised. Available series were then averaged within four groups: housing/collateral, rate/refinancing, labour/income, and financial conditions. For group g ,

$$F_{t,g} = \frac{1}{|\mathcal{G}_g|} \sum_{j \in \mathcal{G}_g} \frac{a_j M_{t,j} - \overline{a_j M_j}}{s(a_j M_j)}, \quad (3.3)$$

where $a_j \in \{-1, 1\}$ sets the adverse direction. The four factors were standardised again before HMM estimation. Equal weighting retained a direct mapping from the source series to each factor.

For the original terminal comparison, the HMM estimation window was January 2014–September 2024. It included a twelve-month margin before the first mortgage observation. The factor transformations, scaler, and HMM parameters were estimated once over this complete window. The causal rolling-origin analysis retained the same four factor definitions but re-estimated their means and standard deviations, the second-stage scaler, and the HMM using macroeconomic observations available through each training cutoff only.

Table 3.3: Macroeconomic transformations and HMM grouping

Factor	Source transformations	Adverse orientation
Housing/collateral	Twelve-month house-price growth	Lower growth
Rate/refinancing	Policy rate; mortgage rate; mortgage–Treasury spread; 10-year rate and monthly change; 10-year minus 2-year spread	Higher rates and spread; lower yield curve
Labour/income	Unemployment and change; claims and change; twelve-month payroll and real-income growth; sentiment and change	Higher unemployment and claims; lower growth and sentiment
Financial conditions	National Financial Conditions Index	Higher index

Inflation remained in the 17-variable observed macroeconomic vector used by the additive baseline. It was not assigned to one of the four retained HMM factor groups. The HMM source set therefore differed from the complete macro-augmented logistic input. The additive baseline received every available transformed macro series, while the HMM received the four grouped factors recorded in the retained HMM artefact.

3.4 Two-State Hidden Markov Model

Let $R_t \in \{0, 1\}$ denote the latent macroeconomic state and let Z_t be the four-dimensional standardised factor vector. The state process followed a first-order Markov chain,

$$\Pr(R_t = k \mid R_{t-1} = j) = p_{jk}, \quad P = \begin{bmatrix} p_{00} & p_{01} \\ p_{10} & p_{11} \end{bmatrix}, \quad (3.4)$$

and the emissions followed state-specific multivariate Gaussian distributions,

$$Z_t \mid R_t = k \sim \mathcal{N}(\mu_k, \Sigma_k), \quad k \in \{0, 1\}. \quad (3.5)$$

Both covariance matrices were unrestricted. The model used two states, full covariance matrices, and a maximum of 500 estimation iterations. The terminal analysis used random

seed 42. The rolling-origin analysis used the fixed candidate seeds 11, 23, 42, 71, and 101; invalid or non-converged fits were excluded, and the converged fit with the greatest training log likelihood was selected without reference to validation or test loan outcomes. Parameters were estimated by the expectation-maximisation procedure for Gaussian HMMs (Baum et al., 1970; Bilmes, 1998).

For a latent state path $R_{1:T}$ and observed factors $Z_{1:T}$, the joint density was

$$f(Z_{1:T}, R_{1:T}; \Theta) = \delta_{R_1} \prod_{t=2}^T p_{R_{t-1}, R_t} \prod_{t=1}^T \phi_4(Z_t; \mu_{R_t}, \Sigma_{R_t}), \quad (3.6)$$

where δ is the initial state distribution and ϕ_4 is the four-dimensional Gaussian density. The expectation step calculated posterior state and transition probabilities under the current parameter values. The maximisation step updated the initial probabilities, transition matrix, state means, and covariance matrices.

The fitted state labels were oriented after estimation. If $\hat{\mu}_{k,g}$ is the fitted mean of adverse-oriented factor g in state k , the state score was

$$s_k = \sum_{g=1}^4 \hat{\mu}_{k,g}. \quad (3.7)$$

The state with the larger s_k was labelled Stress. This rule used all four HMM inputs and did not define stress from the frequency or chronology of the fitted state.

The state with the larger sum of adverse-oriented fitted factor means was labelled Stress. The forward recursion produced

$$\alpha_1(k) \propto \delta_k f_k(Z_1), \quad (3.8)$$

$$\alpha_t(k) \propto f_k(Z_t) \sum_{j=0}^1 \alpha_{t-1}(j) p_{jk}, \quad (3.9)$$

with normalisation at each month. The retained macroeconomic input was

$$q_t = \Pr(R_t = \text{Stress} \mid Z_{1:t}; \hat{\Theta}). \quad (3.10)$$

This was a filtered rather than smoothed probability. It did not condition on factor observations after month t in the state recursion. In the terminal analysis it remained conditional on transformations and HMM parameters estimated over the full January 2014–September 2024 window. In a rolling fold, the fitted transformations, scaler, transition matrix, and emission parameters were held fixed after the training cutoff while the forward recursion processed validation and test observations in chronological order.

The fitted output also contained $\mathbb{I}(q_t \geq 0.5)$, three-, six-, and twelve-month lags of q_t , the monthly change Δq_t , and the duration of the current sequence above 0.5. The primary regime-aware model used only q_t . The hard indicator was used for descriptive regime slices. The smaller dynamic diagnostic used q_t , Δq_t , and duration. The lagged values were retained in the HMM output but were not included in the reported dynamic model.

3.5 Loan-Level PD Models

All PD specifications were regularised logistic models. Numerical inputs and interaction terms were standardised within each fitted pipeline. The borrower-only baseline was

$$\pi_{i,t}^{(B)} = \sigma(\beta_0 + X_{i,t}^\top \beta), \quad (3.11)$$

where $\sigma(u) = (1 + e^{-u})^{-1}$. The macro-augmented baseline added the 17 transformed macroeconomic variables M_t :

$$\pi_{i,t}^{(M)} = \sigma(\beta_0 + X_{i,t}^\top \beta + M_t^\top \gamma). \quad (3.12)$$

The regime-aware model used q_t and selected interactions:

$$\pi_{i,t}^{(R)} = \sigma(\beta_0 + X_{i,t}^\top \beta + \gamma_R q_t + q_t W_{i,t}^\top \theta), \quad (3.13)$$

where $W_{i,t}$ contained credit score, original combined loan-to-value ratio, original debt-to-income ratio, original loan-to-value ratio, original interest rate, current delinquency in months, recent 60-day delinquency within three months, and months since 30-day delinquency. The marginal log-odds coefficient for an interacted variable $W_{i,t,j}$ was $\beta_j + q_t \theta_j$. The specification therefore allowed selected loan-level sensitivities to vary continuously with the estimated stress probability.

Table 3.4: Model hierarchy

Model	Inputs
Borrower-only	$X_{i,t}$
Macro-augmented	$X_{i,t}$ and the 17 transformed macroeconomic variables
	M_t
Stress level only	$X_{i,t}$ and q_t
Regime-aware	$X_{i,t}$, q_t , and $q_t W_{i,t}$
Regime-aware plus macro	$X_{i,t}$, M_t , q_t , and $q_t W_{i,t}$
Dynamic stress	$X_{i,t}$, q_t , $q_t W_{i,t}$, Δq_t , and stress duration

Three additional models separated components of this structure. A stress-level model added q_t to $X_{i,t}$ without interactions. A regime-aware plus raw-macro model combined Equation (3.13) with M_t . A dynamic-stress diagnostic added $\Delta q_t = q_t - q_{t-1}$ and the number of consecutive months for which $q_t \geq 0.5$. These models were evaluated on the same target and chronological samples.

The implementation used stochastic-gradient logistic regression with an L_2 penalty. Candidate penalty values were $\{3 \times 10^{-6}, 10^{-5}, 3 \times 10^{-5}\}$. Each value was evaluated on deterministic samples of 500,000 training and 500,000 validation observations. The value with the lowest validation Brier score was selected, and the model was then fitted to the complete training sample.

For coefficient vector ω and intercept β_0 , estimation minimised a penalised Bernoulli

objective of the form

$$- \sum_{(i,t) \in \mathcal{R}_{\text{train}}} \left[Y_{i,t}^{(12)} \log \pi_{i,t} + (1 - Y_{i,t}^{(12)}) \log(1 - \pi_{i,t}) \right] + \lambda \|\omega\|_2^2. \quad (3.14)$$

The intercept was separate from the penalised feature vector. Standardisation was fitted within the model pipeline. The coefficient artefact therefore recorded effects on the standardised scale. The empirical analysis used the coefficients to define an explicit predictive model rather than to make causal claims.

3.6 Chronological Validation and Calibration

The 117 model months were ordered before splitting. The first 70% formed the training period, the next 15% formed the validation period, and the final 15% formed the test period. This produced January 2015–September 2021 for training, October 2021–March 2023 for validation, and April 2023–September 2024 for testing. Model selection used the validation period. Test outcomes were reserved for final evaluation.

Raw probabilities were calibrated using the complete validation predictions. The primary method was Platt scaling,

$$\hat{\pi}^{\text{Platt}} = \sigma(a + b \logit(\hat{\pi}^{\text{raw}})). \quad (3.15)$$

Intercept-only recalibration and isotonic regression were retained as sensitivity analyses. The fitted calibration maps were applied unchanged to training, validation, and test predictions. All primary comparisons used the Platt-calibrated probability.

The intercept-only map had the form

$$\hat{\pi}^{\text{Int}} = \sigma(c + \logit(\hat{\pi}^{\text{raw}})), \quad (3.16)$$

where c was selected so that the validation mean matched the validation event rate. Isotonic

regression estimated a non-decreasing mapping from raw validation probabilities to validation outcomes, with values outside the fitted range clipped to the fitted boundaries. Neither sensitivity map used test outcomes during fitting.

3.7 Causal Rolling-Origin Robustness Protocol

The final robustness analysis was specified before its results were inspected. It froze the two-state full-covariance HMM, the borrower-only, macro-augmented, and regime-aware logistic specifications, the regularisation candidates, the descriptive threshold $q_t \geq 0.5$, and Platt calibration as the primary probability map. It did not use the stress-level-only, regime-aware plus macro, or dynamic-stress diagnostics from the terminal model ladder, and it did not introduce a new model class.

Three origins used non-overlapping twelve-month test windows. Each test window was preceded by a twelve-month validation window, while the training sample expanded through the month immediately before validation. Table 3.5 defines the dates.

Table 3.5: Causal rolling-origin design

Origin	Training endpoint	Validation window	Test window
1	March 2020	April 2020–March 2021	April 2021–March 2022
2	March 2021	April 2021–March 2022	April 2022–March 2023
3	March 2022	April 2022–March 2023	April 2023–March 2024

At origin o , let T_o be the training endpoint. For every macroeconomic source variable, the adverse-oriented mean and standard deviation in Equation (3.3) were estimated using dates no later than T_o . The four resulting factors were standardised using the same restricted period. Each deterministic HMM initialisation was fitted to that training sequence. The selected transition and emission parameters and both levels of scaling were then frozen through the validation and test windows. This removed the full-window parameter informa-

tion present in the terminal feature path.

Within each fold, the logistic penalty was selected on deterministic samples of at most 500,000 training and 500,000 validation loan-months using validation Brier score. The selected logistic model was refitted to the complete fold-specific training sample. Platt scaling was estimated from complete validation predictions and applied unchanged to the fold test. Isotonic calibration was not considered in the freeze pass because choosing it after inspecting terminal test outcomes would violate the pre-specified calibration rule.

Fold-specific metrics were reported separately. Pooled rolling metrics concatenated the three non-overlapping test windows, giving every retained loan-month one contribution. This pooled sample did not include April–September 2024 and was not numerically identical to the original terminal test. Its purpose was to assess recurrence across consecutive forecast origins rather than replace the original terminal-period description.

Paired uncertainty compared the regime-aware and borrower-only losses on the same loan-months. For metric $m \in \{BS, LL\}$, the monthly loss difference was

$$d_t^{(m)} = \frac{1}{n_t} \sum_{i \in \mathcal{R}_t} \left[\ell_m \left(Y_{i,t}^{(12)}, \hat{\pi}_{i,t}^{(R)} \right) - \ell_m \left(Y_{i,t}^{(12)}, \hat{\pi}_{i,t}^{(B)} \right) \right], \quad (3.17)$$

so a negative value favoured regime-aware. Ten thousand moving-block resamples used consecutive three-month blocks across the 36 ordered test months. Percentile intervals were calculated from the pooled row-weighted loss difference in each resample. This procedure preserved short sequences of common macroeconomic conditions and avoided treating millions of loan-months as independent time observations. With only 36 test months, the interval remained a limited temporal uncertainty diagnostic rather than a general population confidence interval.

3.8 Evaluation Measures

The primary probability measures were the Brier score and log loss. For an evaluation sample $\mathcal{S}$,

$$BS = \frac{1}{|\mathcal{S}|} \sum_{(i,t) \in \mathcal{S}} (\hat{\pi}_{i,t} - Y_{i,t}^{(12)})^2, \quad (3.18)$$

$$LL = -\frac{1}{|\mathcal{S}|} \sum_{(i,t) \in \mathcal{S}} \left[Y_{i,t}^{(12)} \log \hat{\pi}_{i,t} + (1 - Y_{i,t}^{(12)}) \log(1 - \hat{\pi}_{i,t}) \right]. \quad (3.19)$$

Lower values indicate better probability forecasts. The Brier score was also decomposed over ten equal-count probability bins into reliability, resolution, and uncertainty (Brier, 1950; Murphy, 1973; Dimitriadis et al., 2020).

For bin b , let n_b be its size, $\bar{\pi}_b$ its mean prediction, $\bar{y}_b$ its observed event rate, and $\bar{y}$ the overall event rate. The grouped components were

$$\text{Reliability} = \sum_{b=1}^{10} \frac{n_b}{n} (\bar{\pi}_b - \bar{y}_b)^2, \quad (3.20)$$

$$\text{Resolution} = \sum_{b=1}^{10} \frac{n_b}{n} (\bar{y}_b - \bar{y})^2, \quad (3.21)$$

$$\text{Uncertainty} = \bar{y}(1 - \bar{y}). \quad (3.22)$$

The grouped Brier diagnostic equalled reliability minus resolution plus uncertainty. Equal-count bins were constructed by ranking predictions before division into deciles. The grouped value was not expected to equal the ungrouped score exactly.

ROC-AUC and PR-AUC measured discrimination. PR-AUC was reported because the serious-delinquency outcome was rare (Davis and Goadrich, 2006; Saito and Rehmsmeier, 2015). Calibration curves compared mean predicted PD with observed event frequency across ten bins. Annual test-period Brier scores described temporal variation. The monthly portfolio diagnostic compared observed defaults, $D_t = \sum_i Y_{i,t}^{(12)}$, with predicted defaults,

$\hat{D}_t = \sum_i \hat{\pi}_{i,t}$, using mean absolute count error and mean absolute relative error.

Test observations were also divided into ten quantile bins of q_t . Each bin recorded its stress-probability range, event rate, mean prediction, and model metrics. These bins described numerical gradients within the terminal period. They did not supply independent macroeconomic episodes because each observation in every terminal bin had a hard-stress indicator equal to one.

For every rolling fold and split, saturation diagnostics recorded the minimum, mean, maximum, standard deviation, hard-stress share, shares below 0.01 and above 0.99, the share at or beyond 0.001 and 0.999, and mean binary entropy. These diagnostics tested whether causal parameter fitting supplied a useful continuous state probability. The analysis did not change the hard-state threshold or transform posterior probabilities to manufacture intermediate values.

The reported uncertainty analysis resampled months but did not implement a two-way loan-and-month cluster bootstrap. Repeated observations from the same loan and common macroeconomic exposure within month therefore remain sources of dependence. The rolling design and block interval address temporal recurrence more directly than the terminal split, but they do not convert the 6.65 million pooled loan-months into 6.65 million independent macroeconomic observations.

Results

4.1 Sample and Outcome

The eligible panel contained 15,461,471 loan-months from 438,713 loans across the 2015–2024 cohorts. It contained 167,056 twelve-month transitions to 90 or more days past due, an event rate of 1.0805%. The training sample contained 8,179,648 observations with an event rate of 1.5191%. The validation sample contained 3,175,548 observations with an event rate of 0.4982%. The test sample contained 4,106,275 observations and 26,983 events, an event rate of 0.6571%. The variation in event rates across the chronological samples shows that the probability level changed materially over time.

Table 4.1 reports the retained cohort coverage. Earlier cohorts contributed longer histories and more loan-months. The 2023 and 2024 cohorts contributed fewer observations because the model-ready data ended in September 2024 and each included loan-month required twelve subsequent observed months. Cohort event rates were not monotonic. They ranged from 0.4361% for the 2020 cohort to 2.0893% for the 2019 cohort.

The cohort table describes origination groups, while the training, validation, and test samples were divided by observation month. A loan from an early cohort could therefore contribute to more than one chronological split as it aged. No future outcome for a test-period loan-month entered model selection or probability calibration. The lower validation event rate and later increase in the test rate created a direct test of temporal calibration transfer.

Table 4.1: Eligible sample by origination cohort

Cohort	Loans	Loan-months	Events	Event rate
2015	46,574	2,756,432	21,296	0.007726
2016	47,303	2,653,282	24,896	0.009383
2017	47,144	2,077,915	30,191	0.014529
2018	44,005	1,360,031	28,405	0.020886
2019	39,514	1,229,576	25,689	0.020893
2020	42,751	1,744,236	7,606	0.004361
2021	47,277	1,691,649	8,233	0.004867
2022	47,807	1,187,079	13,177	0.011100
2023	46,158	630,299	6,653	0.010555
2024	30,180	130,972	910	0.006948
Total	438,713	15,461,471	167,056	0.010805

4.2 Estimated Macroeconomic Regimes

The HMM used 129 monthly factor observations from January 2014 to September 2024. The fitted stress probability ranged from approximately 1.63×10^{-102} to a value numerically equal to one over that window. Ninety-nine months were below the hard-state threshold of 0.5 and 30 months were above it. Figure 4.1 shows the filtered probability path.

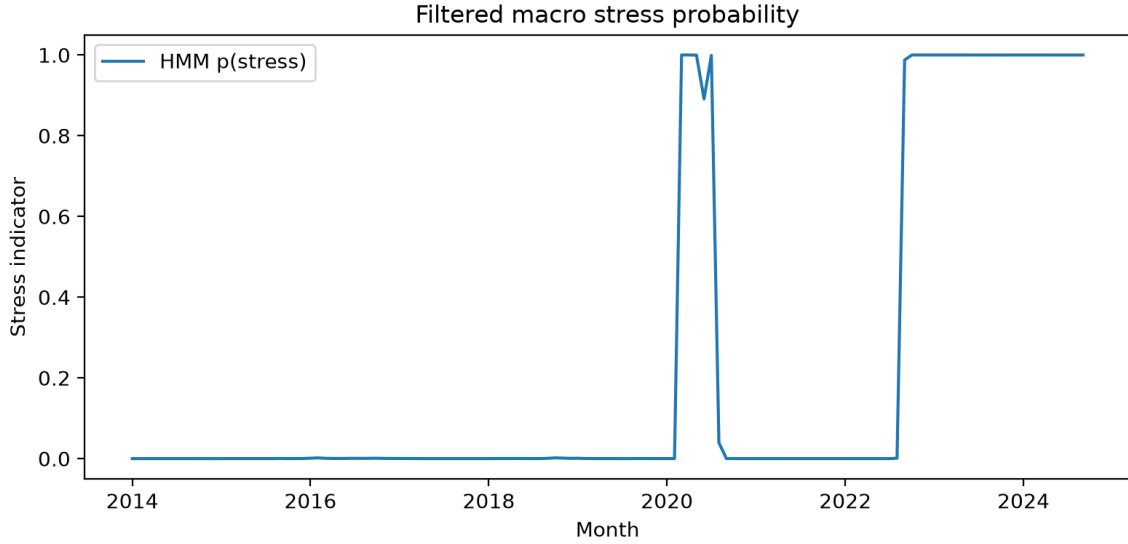


Figure 4.1: Forward-filtered probability of the HMM stress state, January 2014–September 2024. Factor transformations and HMM parameters were estimated over the full displayed window.

The probability was nearly saturated during the test period. It ranged from 0.999903 to a value numerically indistinguishable from one, and every test observation had $q_t \geq 0.5$. The test data therefore did not contain a hard normal-regime comparison. Quantile bins separated only small numerical differences near one. The retained HMM answered the first research question by assigning a persistent stress state throughout the terminal test period, but it provided weak variation for estimating cross-regime test performance.

4.3 Validation Results

The borrower-only model performed best on the validation sample under the primary Platt calibration. Its Brier score was 0.004645 and its log loss was 0.026122. The regime-aware values were 0.004682 and 0.027021. The macro-augmented values were 0.004683 and 0.026850. Borrower-only also had the highest validation ROC-AUC, 0.798634, and PR-AUC, 0.114809. The corresponding regime-aware values were 0.767182 and 0.105560. The validation evidence did not show an advantage for the regime-aware specification.

Table 4.2: Validation performance using Platt-calibrated probabilities

Model	Brier	Log loss	ROC-AUC	PR-AUC
Borrower-only	0.004645	0.026122	0.798634	0.114809
Macro-augmented	0.004683	0.026850	0.775172	0.106811
Stress level only	0.004685	0.026627	0.786294	0.105784
Regime-aware	0.004682	0.027021	0.767182	0.105560
Regime-aware plus macro	0.004722	0.027404	0.756329	0.098194
Dynamic stress	0.004694	0.026656	0.784194	0.098966

The stress-level-only model had lower validation log loss than the regime-aware model but a slightly higher Brier score. The regime-aware plus macro model was weakest on validation Brier and log loss. The dynamic-stress model did not improve on borrower-only. The ranking therefore did not support adding complexity on the basis of the validation period.

Isotonic calibration reduced the validation Brier score to 0.004629 for borrower-only and 0.004647 for regime-aware. These values did not change the validation ordering. Platt scaling

remained the primary calibration method specified in the implementation.

The regularisation choices also differed. Borrower-only, macro-augmented, stress-level-only, regime-aware plus macro, and dynamic stress selected 3×10^{-5} . Regime-aware selected 3×10^{-6} . These values were selected from the same candidate set using deterministic training and validation samples. The later test ranking was not used to choose them.

4.4 Held-Out Test Results

Table 4.3 reports the primary Platt-calibrated test results. The regime-aware model had the lowest Brier score and log loss and the highest ROC-AUC and PR-AUC. Relative to borrower-only, its Brier score fell from 0.006055 to 0.006015, a reduction of 0.66%. Its log loss fell from 0.033106 to 0.032945, a reduction of 0.48%. ROC-AUC increased by 0.005276 and PR-AUC increased by 0.011736.

Table 4.3: Held-out test performance using Platt-calibrated probabilities

Model	Mean PD	Brier	Log loss	ROC-AUC	PR-AUC
Borrower-only	0.010747	0.006055	0.033106	0.821979	0.148361
Macro-augmented	0.009743	0.006078	0.033453	0.805984	0.133367
Stress level only	0.013348	0.006089	0.034349	0.825747	0.150199
Regime-aware	0.010802	0.006015	0.032945	0.827256	0.160097
Regime-aware plus macro	0.011313	0.006105	0.034187	0.804293	0.131448
Dynamic stress	0.000687	0.006472	0.045723	0.745641	0.077918

Note: $n = 4,106,275$ loan-months; 26,983 events; event rate 0.006571. Lower Brier score and log loss are better. Higher ROC-AUC and PR-AUC are better.

The nested specifications qualified this result. Adding q_t without interactions produced a Brier score of 0.006089 and log loss of 0.034349. Adding the full macro vector to the regime-aware model produced 0.006105 and 0.034187. The dynamic-stress model substantially underpredicted the test event rate and performed worst on all four reported measures. The held-out advantage was specific to the selected regime-aware interaction structure.

The test ranking differed from the validation ranking. This limits the strength of the conclusion. The evidence supports a modest terminal-period improvement in pooled loan-

level performance. It does not show that the improvement was stable across the validation and test periods.

The absolute test differences were small relative to the scale of the scores. The Brier difference between regime-aware and borrower-only was 0.0000397. The log-loss difference was 0.0001603. The report tables do not contain uncertainty intervals for these paired differences. The observed ordering is exact for the retained predictions, but its sampling uncertainty was not estimated.

4.5 Calibration

The test event rate was 0.006571. The mean Platt-calibrated PD was 0.010747 for borrower-only, 0.009743 for macro-augmented, and 0.010802 for regime-aware. Each core model over-predicted the mean event rate. Figure 4.2 shows that validation-based calibration did not remove this temporal shift.

The ten-bin Brier decomposition gave reliability of 0.0000293 and resolution of 0.0001017 for the regime-aware model. Borrower-only had reliability of 0.0000549 and resolution of 0.0001006. The regime-aware grouped diagnostic therefore had lower reliability error and similar resolution. The macro-augmented model had lower grouped reliability, 0.0000247, but worse ungrouped Brier score and discrimination. The grouped decomposition did not reproduce the complete model ordering.

Table 4.4 compares calibration maps for the two principal models. Isotonic calibration slightly improved the regime-aware Brier score and log loss relative to Platt. It worsened both measures for borrower-only. This was a sensitivity result because the primary probability column used Platt scaling.

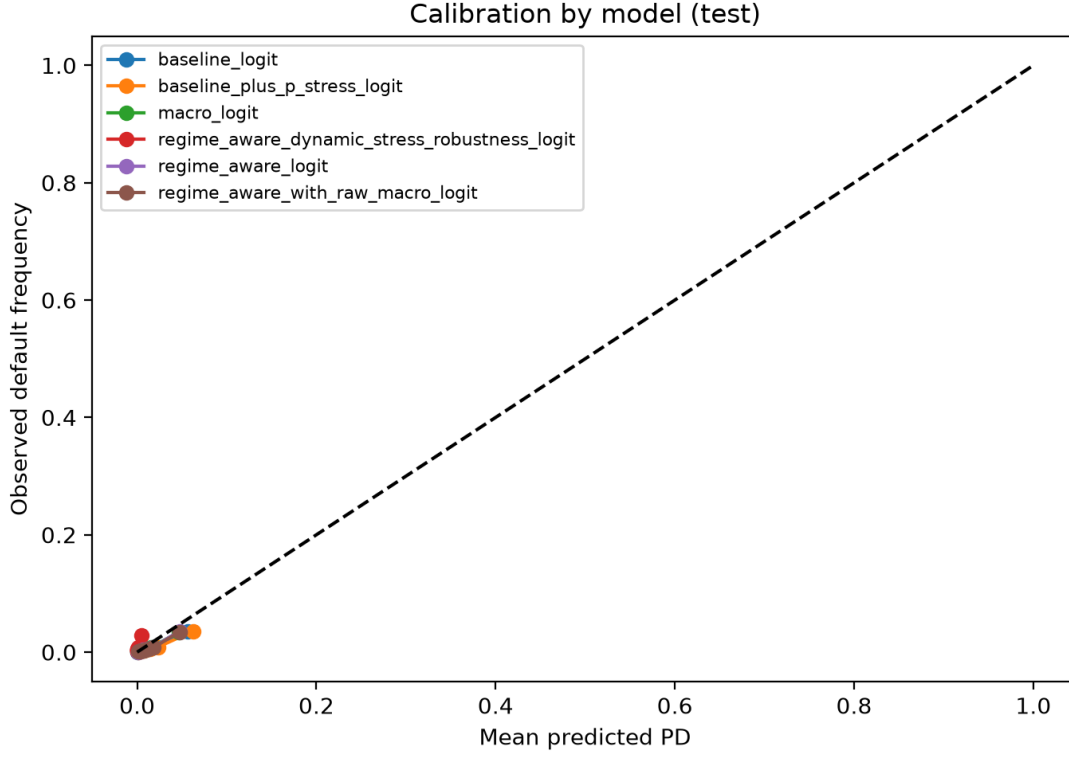


Figure 4.2: Calibration curves for the held-out test period. Points compare mean predicted PD with observed event frequency in equal-count probability bins.

Table 4.4: Calibration sensitivity for the principal test models

Model	Calibration	Mean PD	Brier	Log loss
Borrower-only	Raw	0.035379	0.008941	0.052552
Borrower-only	Intercept-only	0.011309	0.006120	0.033404
Borrower-only	Isotonic	0.011024	0.006125	0.033322
Borrower-only	Platt	0.010747	0.006055	0.033106
Regime-aware	Raw	0.063093	0.015897	0.084550
Regime-aware	Intercept-only	0.013791	0.006673	0.035555
Regime-aware	Isotonic	0.011020	0.006007	0.032811
Regime-aware	Platt	0.010802	0.006015	0.032945

4.6 Temporal, Portfolio, and Discrimination Diagnostics

The regime-aware model had a lower Brier score than borrower-only in both test-year groups. In 2023, the scores were 0.005432 and 0.005445. In 2024, they were 0.006546 and 0.006609. Both groups belonged to the same hard-stress classification and therefore did not constitute separate regime tests.

Monthly count calibration favoured borrower-only. Across the 18 test months, its mean absolute predicted-default-count error was 952.5, compared with 965.3 for regime-aware and 1,212.3 for macro-augmented. Mean absolute relative error was 0.6097 for borrower-only, 0.6367 for regime-aware, and 0.9016 for macro-augmented. The pooled Brier improvement did not translate into a lower average monthly count error.

Table 4.5: Monthly default-count error over the 18-month test period

Model	Mean absolute count error	Mean absolute relative error
Borrower-only	952.5	0.6097
Regime-aware	965.3	0.6367
Macro-augmented	1,212.3	0.9016
Dynamic stress	1,342.3	0.8751
Regime-aware plus macro	1,347.5	0.9957
Stress level only	1,546.0	1.0075

Mean predicted PD exceeded the event rate for the three core models, so the positive monthly count errors were consistent with the pooled mean-calibration result. Borrower-only produced the smallest average error even though regime-aware allocated risk more effectively across individual loan-months according to the pooled Brier and ranking measures. This difference separates cross-sectional allocation from portfolio-level calibration.

Figure 4.3 shows test ROC curves. The regime-aware ROC-AUC of 0.827256 exceeded the borrower-only value of 0.821979 and the macro-augmented value of 0.805984. PR-AUC showed the same ordering. This indicates better held-out ranking for the regime-aware model

in the terminal period. It does not resolve the mean overprediction or the weaker validation results.

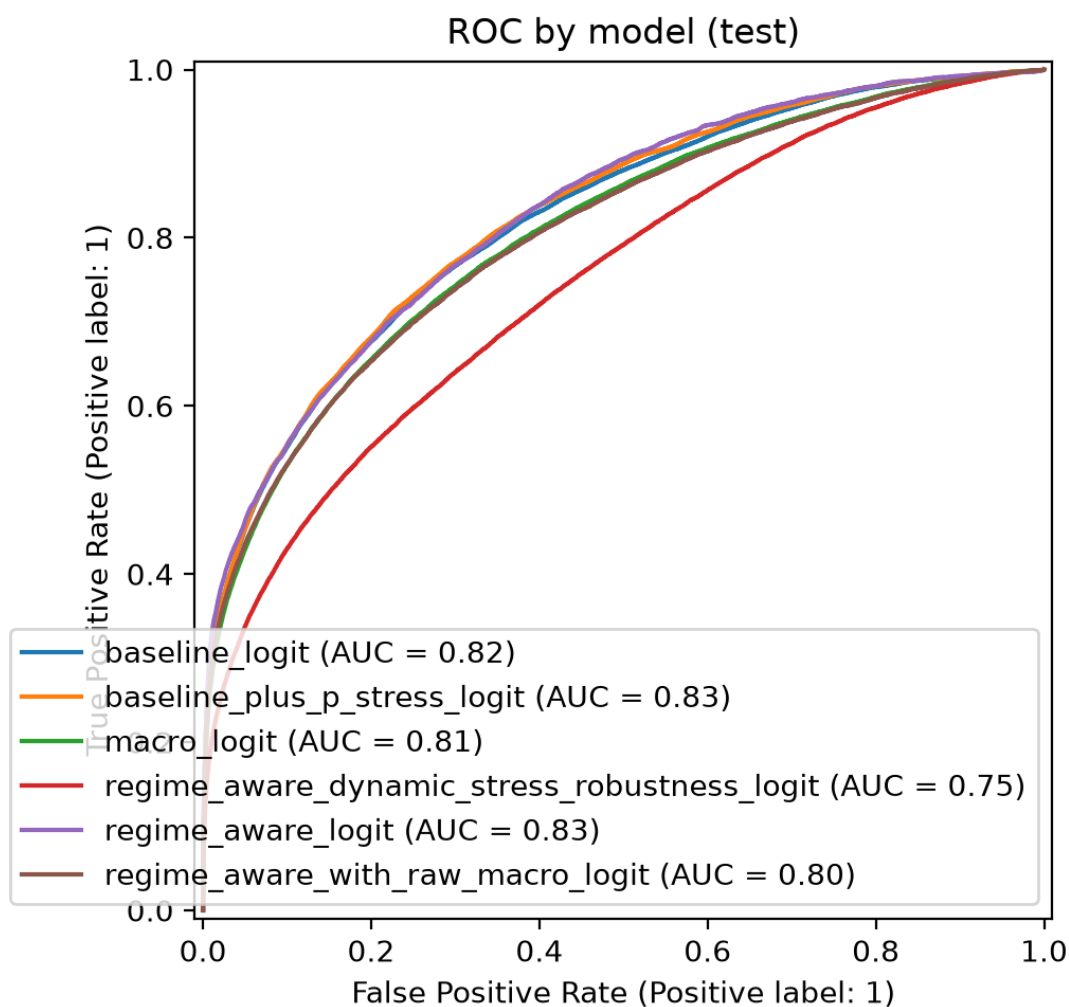


Figure 4.3: Receiver operating characteristic curves for the held-out test period.

Figure 4.4 reports the precision–recall curves. The test event rate of 0.006571 is the no-skill prevalence reference for average precision. The regime-aware PR-AUC was 0.160097, compared with 0.148361 for borrower-only and 0.133367 for macro-augmented. The ordering matched ROC-AUC, but the PR scale made the rare-event context explicit.

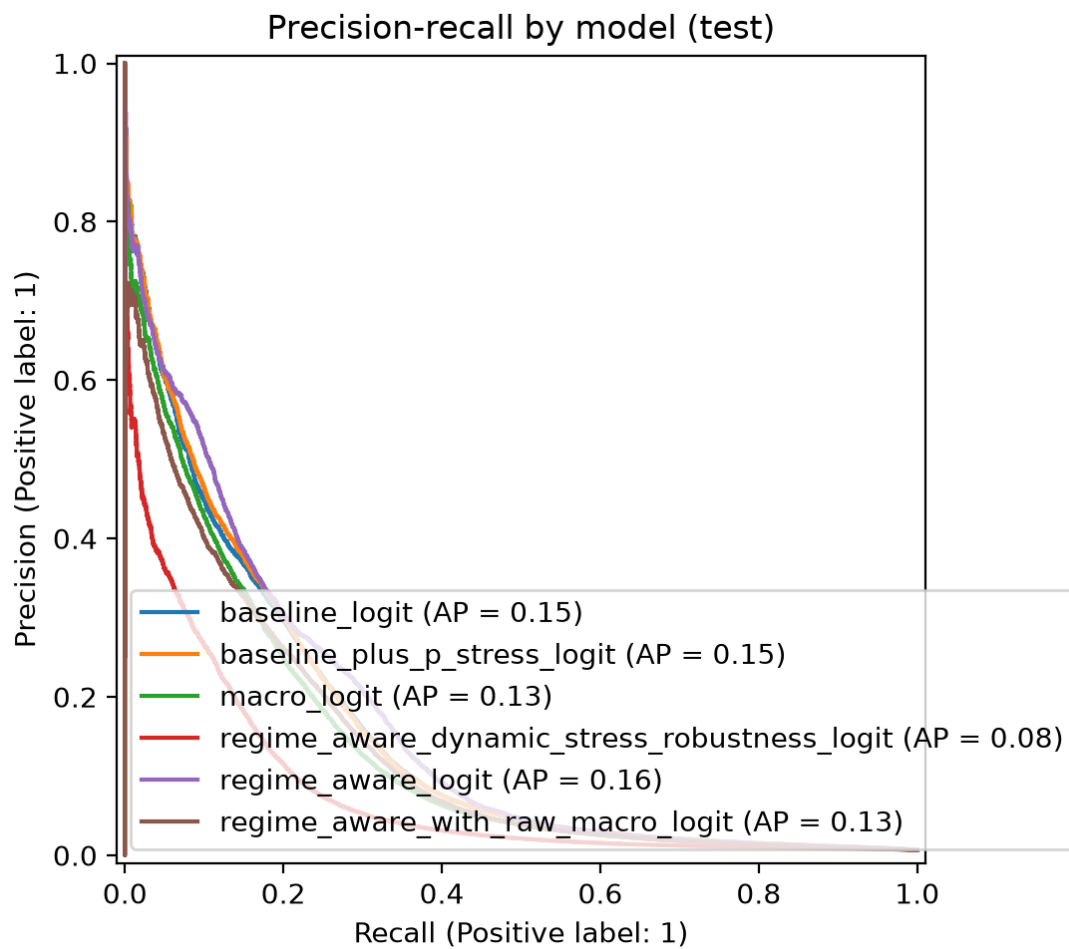


Figure 4.4: Precision-recall curves for the held-out test period.

4.7 Causal Rolling-Origin HMM Diagnostics

The rolling analysis fitted five deterministic HMM starts at each origin and selected training likelihood without using validation or test loan outcomes. Selected seeds were 101, 71, and 71. At the first origin, seeds 23 and 71 produced invalid transition matrices because one state had no observed outgoing transition; these starts were excluded. The remaining three first-origin starts and all five starts at each later origin converged to valid fitted transition matrices. This result illustrates why deterministic multiple starts were retained even though the number of states and covariance structure were fixed.

Table 4.6 reports stress diagnostics for the three test windows. Causal fitting changed the hard-state composition relative to the terminal full-window feature path: the first rolling test contained one hard-stress month, the second contained twelve, and the third contained eight. It did not produce a well-spread continuous probability. All first- and second-origin test months and eleven of twelve third-origin months had $q_t \leq 0.001$ or $q_t \geq 0.999$.

Table 4.6: Rolling-test stress-probability diagnostics

Test window	Minimum	Mean	Maximum	Hard share	Extreme share	Mean entropy
2021–2022	4.42×10^{-144}	0.0833	1.0000	0.0833	1.0000	8.99×10^{-40}
2022–2023	1.0000	1.0000	1.0000	1.0000	1.0000	0.0000
2023–2024	9.01×10^{-28}	0.6528	1.0000	0.6667	0.9167	0.0376

Note: “Extreme share” is the share of months with $q_t \leq 0.001$ or $q_t \geq 0.999$. Binary entropy is zero at probabilities zero and one and largest at 0.5.

The rolling path therefore supplied hard-state variation across the combined 36 months, but almost no within-state posterior uncertainty. The second origin was entirely stress, while the first was almost entirely normal under its independently fitted parameters. These origin-specific labels were oriented using the same adverse-factor rule, yet their absolute posterior values remained sensitive to the fitted window. The HMM result is informative as a period classification and model input, but it is not evidence of a stable smooth stress gradient.

4.8 Causal Rolling-Origin Model Performance

Table 4.7 reports the three core models in each rolling test. Regime-aware improved all four reported measures in the first origin. Its Brier score was 0.005087 compared with 0.005559 for borrower-only, and its log loss was 0.031512 compared with 0.036308. This ordering reversed in the next two tests. In April 2022–March 2023, borrower-only had Brier 0.004788 compared with 0.005004 for regime-aware. In April 2023–March 2024, the respective values were 0.005794 and 0.006145.

Table 4.7: Causal rolling-origin test performance

Test window	Model	Brier	Log loss	ROC-AUC	PR-AUC
2021–2022	Borrower-only	0.005559	0.036308	0.738610	0.095609
	Macro-augmented	0.049712	0.223475	0.615857	0.018066
	Regime-aware	0.005087	0.031512	0.759232	0.101768
2022–2023	Borrower-only	0.004788	0.027028	0.792048	0.109966
	Macro-augmented	0.009824	0.057854	0.732330	0.088417
	Regime-aware	0.005004	0.031107	0.729484	0.080261
2023–2024	Borrower-only	0.005794	0.031987	0.817696	0.130671
	Macro-augmented	0.005764	0.033132	0.811158	0.137750
	Regime-aware	0.006145	0.036403	0.774133	0.091500

The fold pattern was not a weak alternation around an unchanged ranking. In the first origin, regime-aware exceeded borrower-only ROC-AUC by 0.020622 and PR-AUC by 0.006159. In the second, it was lower by 0.062564 and 0.029705. In the third, it was lower by 0.043563 and 0.039171. The advantage was therefore concentrated in one period rather than recurring across origins.

The additive macro model was especially unstable. Its validation event rate before the first test was 2.2800%, while the following test event rate was 0.5098%. The Platt map estimated in that validation period transferred poorly: mean macro-model PD in the first test was 18.6756%. The macro model also overpredicted in the second test. Its poor pooled result should be interpreted as temporal calibration failure of the retained additive specification, not as evidence that macroeconomic information is generally uninformative.

4.9 Pooled Rolling Performance and Calibration

The three non-overlapping test windows contained 6,650,858 loan-months and 36,833 events, an event rate of 0.005538. Table 4.8 reports pooled results. Borrower-only had the lowest Brier score and log loss and the highest ROC-AUC and PR-AUC. Regime-aware overpredicted the mean event rate by a larger amount and did not retain the terminal-split ranking advantage.

Table 4.8: Pooled causal rolling-test performance

Model	Mean PD	Brier	Log loss	ROC-AUC	PR-AUC
Borrower-only	0.011004	0.005394	0.031469	0.784796	0.103932
Macro-augmented	0.068030	0.018795	0.091968	0.689456	0.044764
Regime-aware	0.012392	0.005481	0.033326	0.753599	0.089320

Note: pooled test event rate 0.005538. The pooled sample concatenates the three non-overlapping rolling tests and differs from the original April 2023–September 2024 terminal test.

Monthly portfolio calibration gave the same primary ordering. Mean absolute predicted-default-count error across the 36 rolling-test months was 1,009.7 for borrower-only, 1,266.3 for regime-aware, and 11,647.0 for macro-augmented. The corresponding mean count errors were positive, so overprediction rather than offsetting monthly under- and overprediction explained the borrower-only and regime-aware absolute errors. Regime-aware had lower monthly Brier and log-loss contributions in 15 of the 36 months.

The pooled decile calibration tables in Appendix C show that both principal models overpredicted across much of the probability range. The mean PD was approximately twice the pooled event rate for borrower-only and more than twice for regime-aware. The causal rolling pass therefore strengthened, rather than removed, the calibration-in-the-large limitation observed in the terminal split.

4.10 Paired Temporal Uncertainty

Table 4.9 reports the three-month moving-block results. Positive differences favour borrower-only. The regime-aware-minus-baseline Brier difference was 0.0000869, with a 95% interval from -0.0001558 to 0.0002714 . The log-loss difference was 0.0018573, with an interval from -0.0008756 to 0.0040205 . Both intervals included zero.

Table 4.9: Paired moving-block uncertainty for pooled rolling tests

Loss	Regime-aware minus borrower-only	95% interval	Bootstrap probability regime-aware is better
Brier	0.0000869	$[-0.0001558, 0.0002714]$	0.2518
Log loss	0.0018573	$[-0.0008756, 0.0040205]$	0.0876

The intervals do not prove equivalence. They show that 36 months do not support a precise recurring difference after short temporal blocks are respected. The point estimates favour borrower-only, while resampled sequences include periods in which regime-aware is better. This result is consistent with the fold-specific reversal and is not consistent with a universal regime-aware gain.

4.11 Answers to the Research Questions

The first research question asked what stress probabilities the two-state HMM assigned. The terminal HMM assigned every April 2023–September 2024 month to hard stress and produced probabilities above 0.999902. Causal rolling fits supplied both hard states across the combined tests, but 35 of 36 test months remained at or beyond 0.001 and 0.999. The answer is therefore that the fitted HMM identified persistent, sharply separated periods, but its posterior was too saturated to support a stable continuous cross-regime gradient.

The second question asked how the stress probability and interactions changed the PD specification. Equation (3.13) added a common q_t term and allowed eight selected borrower, leverage, interest-rate, and payment-history sensitivities to vary with q_t . The structure

remained a regularised logistic model and excluded the raw macroeconomic vector. It provided an explicit and inspectable route by which fitted macroeconomic conditions could alter loan-level log odds. The rolling results show that structural interpretability did not ensure temporal predictive stability.

The third question asked how the regime-aware model compared with borrower-only and macro-augmented baselines. In the original terminal test, regime-aware was best on Brier, log loss, ROC-AUC, and PR-AUC, although borrower-only was best on validation and monthly count error. Under causal rolling evaluation, regime-aware won the first origin, lost the next two, and was worse than borrower-only on pooled proper scores, discrimination, and monthly count calibration. The answer is mixed and calibration-first: regime-aware can add period-specific predictive value, but borrower-only was the more stable pooled forecasting model.

4.12 Overall Interpretation

The original terminal result remains valid for the predictions and dates on which it was calculated. It is not the final general comparison. The causal rolling pass removed full-window information from factor scaling and HMM parameter estimation, tested three successive windows, and added temporally blocked paired uncertainty. Under that stricter design, the regime-aware advantage did not recur.

The final empirical conclusion is therefore narrower than a model-win claim. The regime-aware logistic model is an interpretable, period-dependent diagnostic: it states how selected loan sensitivities are allowed to change with a fitted latent macroeconomic condition, and it improved forecasts in one rolling origin and the terminal split. Borrower-only logistic regression is stronger on pooled causal Brier score, log loss, ROC-AUC, PR-AUC, and monthly count calibration. The evidence does not support universal predictive superiority for regime awareness.

Discussion

5.1 Interpretation of the Combined Evidence

This study began with a plausible forecasting proposition: a compact latent macroeconomic state might improve loan-level default probabilities by changing both the common level of risk and selected borrower sensitivities. The terminal split initially supported that proposition. The regime-aware model had the lowest Brier score and log loss and the highest ROC-AUC and PR-AUC in April 2023–September 2024. Those results remain correct for that fitted pipeline and period. They are not, however, sufficient evidence that regime awareness produces a recurring forecasting advantage.

The causal rolling-origin analysis changes the weight assigned to the terminal result. Each origin estimated macroeconomic transformations, scaling parameters, the HMM, logistic coefficients, regularisation strength, and Platt calibration without test-period outcomes. It then repeated the comparison over three successive and non-overlapping twelve-month tests. Regime-aware was best in the first test and worse than borrower-only in the next two. When the 36 test months were pooled, borrower-only was better on both proper scores, both discrimination measures, and monthly default-count calibration. The point estimates therefore support borrower-only as the more stable forecasting specification in this dataset.

The paired moving-block analysis adds an important qualification. Its intervals included zero for both the Brier and log-loss differences. This does not overturn the pooled ordering, and it does not establish equivalence. It indicates that the effective temporal evidence is

much smaller than the count of loan-month rows might suggest. Millions of loans share only 36 rolling-test calendar months. Resampling short monthly blocks accordingly produced sequences in which the period-specific regime-aware advantage received different weight. The defensible result is a pooled advantage for borrower-only accompanied by uncertainty about the exact magnitude of the difference.

The combined evidence is thus more informative than either evaluation alone. The terminal split demonstrates that the regime-aware structure can be useful in a particular period. The rolling analysis demonstrates that this usefulness did not persist automatically when the forecast origin moved. Together they support a period-dependent diagnostic interpretation rather than a universal model-win claim.

5.2 A Calibration-First Reading

Probability forecasts should first be assessed as probabilities. In the rolling tests, borrower-only had a pooled Brier score of 0.005394 and log loss of 0.031469, compared with 0.005481 and 0.033326 for regime-aware. These proper scores favour borrower-only. The ordering is reinforced by monthly count calibration: borrower-only had a mean absolute default-count error of 1,009.7, while regime-aware had 1,266.3. Regime-aware had lower monthly loss in only 15 of 36 months.

Neither principal model was ideally calibrated in the large. The pooled event rate was 0.005538, while mean predicted PD was 0.011004 for borrower-only and 0.012392 for regime-aware. The validation-fitted Platt maps therefore did not fully transport to the following test periods. This common overprediction matters operationally. A model can rank individual loans effectively while overstating expected portfolio defaults, capital needs, or the volume of accounts entering an intervention queue. The monthly aggregation makes that consequence visible.

The terminal split illustrated a related distinction. Regime-aware improved pooled Brier

score and ranking there, but borrower-only produced the smaller average monthly count error. The apparent inconsistency is only an aggregation difference. A probability model can reallocate risk more effectively among loans and still predict too many events in aggregate. For this reason, a small improvement in a pooled loan-level score should not be presented as a complete calibration success.

ROC-AUC and PR-AUC remain useful because defaults are rare and ordering matters for review or intervention. Yet they do not repair probability levels. In the pooled rolling tests, borrower-only was also stronger on these measures, with ROC-AUC 0.784796 and PR-AUC 0.103932 compared with 0.753599 and 0.089320 for regime-aware. The agreement of proper-score, ranking, and count-calibration evidence makes the primary rolling conclusion clearer than it would be if the metrics disagreed.

The additive macro model provides a warning about temporal calibration. Its first-test mean PD of 18.6756% followed a validation period with a much higher event rate than the subsequent test. A validation-fitted monotone calibration map could not guarantee transport when the economic and portfolio distribution changed. This failure concerns the retained additive specification and its calibration under the observed shift. It is not evidence that macroeconomic information is universally unhelpful.

5.3 Meaning and Limits of the Regime Signal

The two-state HMM fulfilled its intended compression role. It mapped four adverse-oriented macroeconomic factors into a filtered probability, and its transition structure represented persistence rather than classifying each month independently. The regime-aware logit then supplied an explicit pathway through which the common state could shift loan-level log odds and alter eight selected sensitivities. This is substantively more interpretable than an unrestricted interaction search.

The estimated posterior nevertheless approached a hard classifier. In the terminal test

every month had stress probability above 0.999902. Under causal rolling fitting, 35 of 36 test months had probability at or beyond 0.001 and 0.999. The second rolling test was entirely stress, and the first contained only one hard-stress month. Those values leave little support for interpreting a smooth change from normal to stress inside a fold.

Saturation does not make the model scores invalid. Predictions can still differ across loans because borrower characteristics vary and because interaction coefficients were estimated before the test. It does restrict the economic claim. The analysis cannot reliably estimate how forecast performance changes across intermediate stress probabilities when almost no intermediate probabilities occur. Hard-state comparisons within an all-stress or almost-all-normal test are similarly weak. The HMM is best read here as an origin-specific period classification and an interpretable model input, not as a precisely measured continuous stress intensity.

The independently fitted rolling HMMs also show parameter sensitivity. State labels were consistently oriented by the adverse-factor rule, but their test-state composition changed with the training window. This is expected in a latent-variable model: numeric labels have no inherent meaning, and emission and transition estimates depend on the history available at the origin. Deterministic multiple starts reduced optimisation dependence, while training likelihood selected among valid starts without test information. It did not remove sampling uncertainty from a model estimated on a relatively short monthly series.

The saturated posterior is therefore both a result and a limitation. It indicates that the fitted Gaussian states were sharply separated under the selected factors. It also means that a hard threshold of 0.5 contributed little additional information over most evaluation windows. A richer state model, alternative covariance restriction, or different factor construction might produce a smoother probability, but choosing such variants after observing these test results would constitute new model search. They are appropriate subjects for future research rather than retrospective changes to the frozen thesis comparison.

5.4 Interpretability and Predictive Stability

Interpretability is a property of model structure, not proof of forecast quality. The regime-aware equation identifies the borrower variables whose log-odds sensitivities are permitted to change with q_t . It also separates that channel from a common stress-level term. This makes the source of a probability change inspectable and provides a coherent diagnostic language for discussing borrower vulnerability under fitted adverse conditions.

The rolling results show why transparency and stability must be reported separately. A model can have a clear equation and still transport poorly when the macroeconomic state, portfolio composition, event prevalence, or calibration relationship changes. Conversely, the simpler borrower-only model can omit an explicit macroeconomic channel and still be the stronger pooled forecast because behavioural and contract variables already capture substantial cross-sectional risk.

The regime-aware model should consequently not replace borrower-only solely because it has a richer economic interpretation. A defensible use is parallel monitoring: borrower-only supplies the primary probability forecast, while regime-aware supplies an interpretable sensitivity diagnostic and a challenger prediction. Material divergence between them can prompt review of macroeconomic conditions, calibration drift, and borrower segments. That use respects the observed period dependence rather than hiding it behind one pooled terminal score.

5.5 Relationship to Existing Work

The results are consistent with the broad literature in a qualified way. Hidden-state and regime-switching credit models establish that persistent economic conditions can be represented explicitly in migration, intensity, and structural-risk settings (Bangia et al., 2002; Yu et al., 2018; Milidonis and Chisholm, 2024). Loan-level macroeconomic studies likewise provide reasons to expect unemployment, interest rates, house prices, and related conditions

to affect credit outcomes (Bellotti and Crook, 2009, 2013). The present study confirms that an interpretable regime channel can add predictive value in some periods.

It also supplies a stricter caution. A terminal holdout can favour a regime-aware model even when successive causal origins do not. Full-window scaling and HMM estimation answer a narrower retrospective parameter-estimation question than origin-specific fitting. By moving every upstream transformation inside the training window and retaining a fixed model family, the rolling pass tests whether the apparent gain survives a forecast-like sequence. In this study it did not recur.

This conclusion does not contradict the conceptual value of regime switching. It distinguishes a mechanism from a performance guarantee. A latent state can be economically meaningful and diagnostically useful without producing lower forecast loss in every period. The result also accords with the forecasting principle that model comparisons depend on horizon, sample, information set, calibration method, and evaluation dates. Claims should remain conditional on those choices.

No unreviewed preprint is used as a core authority in this thesis. In particular, a recent preprint describing a different gradient-boosting and regime-specific calibration system was excluded from the bibliography pending independent verification and supervisor guidance. Its model class and uniformly favourable reported outcome do not provide a like-for-like benchmark for the fixed logistic comparison conducted here.

5.6 Practical Implications

For portfolio forecasting, the evidence favours borrower-only logistic regression as the primary model among the three frozen candidates. It delivered the best pooled causal proper scores, discrimination, and monthly count calibration. This recommendation is comparative, not absolute: its mean PD still exceeded the rolling event rate, so deployment would require continuing calibration surveillance.

The regime-aware model remains useful in three narrower roles. First, it can act as a challenger that indicates whether a macro-conditioned specification is beginning to outperform the primary model. Second, its interaction structure can support qualitative review of which loan characteristics are allowed to become more or less influential under the fitted state. Third, the HMM path can provide a compact descriptive indicator for scenario discussion. None of these roles requires claiming that the model is universally more accurate.

Operational use should preserve the information boundaries of the rolling study. Scaling, HMM parameters, logistic coefficients, and calibration maps should be refitted on a documented schedule using only information available at the forecast date. Monitoring should include Brier score, log loss, mean predicted PD versus event rate, grouped calibration, monthly expected versus realised default counts, and both ROC-AUC and PR-AUC. Large changes in the selected HMM start, state composition, or posterior entropy should be treated as diagnostic events rather than silently absorbed.

Threshold-based credit decisions were outside the scope of this study. The reported metrics evaluate probability forecasts and rankings, not approval, pricing, collection, or capital thresholds. Any decision threshold would require an explicit loss function, capacity constraint, fairness and governance review, and validation on the intended decision population.

5.7 External Validity and African Context

The empirical data describe United States mortgages, not African consumer or mortgage portfolios. Freddie Mac loans are affected by United States underwriting conventions, servicing rules, interest-rate transmission, labour markets, housing markets, and public data infrastructure. The FRED variables and their transformations likewise encode a United States macroeconomic environment. Coefficients, event rates, regime dates, and calibration maps cannot be transferred directly to an African lender.

The relevance to an African research audience is therefore methodological rather than

geographic. The study offers a transparent pipeline for combining loan-level behavioural data with a causal filtered macroeconomic state, and it demonstrates why calibration-first rolling evaluation can overturn a favourable terminal result. This lesson is pertinent wherever macroeconomic variation is substantial and usable credit histories are temporally structured. It should be presented as a validation framework and cautionary empirical case, not as evidence about African borrower behaviour.

5.8 Limitations

First, the loan data are sample files rather than the full Freddie Mac population, and the study considers one twelve-month 90-DPD transition definition. Results may differ for another product, event threshold, horizon, cohort range, or risk-set rule. Second, current-vintage FRED observations were aligned historically. The design prevents future dates from entering rolling transformations but does not reconstruct the real-time vintage that an analyst would have observed at every origin.

Third, the macroeconomic effective sample is monthly and short relative to the loan-month panel. HMM and block-uncertainty conclusions are consequently supported by far fewer independent temporal units than the raw observation count suggests. Fourth, the Gaussian two-state full-covariance HMM is deliberately fixed and parsimonious. Posterior saturation limits smooth-regime interpretation, while alternative state counts or emission families were not searched after test inspection.

Fifth, calibration was estimated on the immediately preceding validation block and applied unchanged to each test. This enforces honest separation but exposes the forecasts to prevalence and relationship shifts. Platt scaling was retained as the primary method by protocol; isotonic results from the terminal sensitivity analysis were not used to choose the rolling method. Sixth, the paired moving-block intervals operate on monthly average losses. They preserve short temporal clustering but do not jointly resample loans and months or

quantify every source of parameter uncertainty.

Finally, the study is predictive rather than causal. Coefficients and interactions are conditional associations in a regularised model. The HMM state is a fitted statistical summary, not an observed economic treatment. These limits do not remove the comparative evidence; they determine how narrowly it should be stated.

Conclusion and Recommendations

6.1 Conclusion

This thesis developed and evaluated an interpretable regime-aware model for twelve-month loan-level mortgage probability of default. The empirical pipeline joined Freddie Mac origination and monthly performance records to transformed FRED macroeconomic series, constructed a forward-looking 90-DPD outcome for eligible loan-months, and compressed four adverse-oriented macroeconomic factors into a filtered probability from a two-state Gaussian HMM. A regularised logistic model used that probability as a common level term and interacted it with selected borrower, leverage, interest-rate, and payment-history features. Borrower-only and additive macroeconomic logistic models supplied the principal comparisons, and validation-fitted Platt scaling produced the primary test probabilities.

The first research question concerned the macroeconomic regimes identified by the HMM. The terminal fit assigned the entire April 2023–September 2024 test period to hard stress with probabilities above 0.999902. The causal rolling fits produced both hard states across 36 months, but 35 months still had probabilities at or beyond 0.001 and 0.999. The HMM therefore identified persistent and sharply separated fitted periods. It did not provide enough intermediate posterior variation to justify a smooth stress-gradient interpretation or reliable within-fold hard-regime comparisons.

The second question concerned how regime information changed the PD specification. The final equation added q_t to the loan-level logit and allowed eight prespecified or deter-

ministically selected borrower and payment-history effects to vary with it. This structure created an explicit, auditable macro-to-credit pathway without retaining the full correlated macroeconomic vector in the main regime-aware model. Its interpretation is conditional and predictive: it identifies which sensitivities the model permits to vary, not causal effects of economic stress.

The third question concerned comparative performance. In the original terminal split, regime-aware achieved modestly better proper scores and discrimination than borrower-only. That result was accompanied by weaker validation performance, worse monthly count calibration, and a saturated terminal state. The predefined causal rolling analysis therefore became the more authoritative robustness comparison. Regime-aware won the first rolling origin and lost the next two. Across 6,650,858 pooled rolling-test loan-months, borrower-only had the lower Brier score and log loss, the higher ROC-AUC and PR-AUC, and the smaller monthly default-count error. Paired three-month moving-block intervals included zero, showing that the exact difference is imprecise across only 36 months, while the point estimates and metric ordering favour borrower-only.

The final conclusion is deliberately calibration-first. The regime-aware model is an interpretable, period-dependent diagnostic, not a universally more accurate forecasting model. Borrower-only logistic regression is stronger in pooled causal proper scores, discrimination, and monthly count calibration in this study. Both principal models overpredicted the pooled rolling event rate, so even the preferred comparative model requires ongoing calibration monitoring.

6.2 Contributions

The study makes four bounded contributions. First, it defines a reproducible loan-month estimand whose outcome, horizon, risk set, and information timing are explicit. Second, it links a filtered latent macroeconomic state to loan-level PD through a small and interpretable

interaction structure. Third, it evaluates probabilities using proper scores, ranking measures, grouped and mean calibration, and monthly expected-default counts rather than relying on discrimination alone. Fourth, it demonstrates empirically that a favourable terminal result can reverse under causal rolling-origin fitting when upstream macroeconomic transformations and HMM parameters are also restricted to training information.

The fourth contribution is central. The stricter evaluation does not merely add another table; it changes the strength of the conclusion. It prevents a full-window latent-state estimate from being treated as if it were generated in real time, and it makes temporal instability visible without introducing new models or selecting a calibrator from test performance. The mixed result is scientifically useful because it defines both the potential and the boundary of the proposed architecture.

6.3 Recommendations

For the frozen empirical comparison, borrower-only logistic regression should be reported as the primary forecasting model. The regime-aware specification should be retained as an interpretable challenger and diagnostic. The additive macroeconomic model should remain a baseline demonstrating the instability of the tested direct specification; its poor rolling performance should not be generalised to all uses of macroeconomic variables.

Any operational implementation should refit transformations, scaling, HMM parameters, logistic coefficients, and calibration maps using only information available at the forecast origin. Monitoring should report calibration-in-the-large, Brier score, log loss, grouped calibration, monthly predicted and observed default counts, ROC-AUC, and PR-AUC. Because the HMM posterior was saturated, monitoring should also include the share of extreme probabilities, hard-state composition, entropy, transition matrices, and the selected deterministic initialisation.

Future research should prioritise external and temporal validation rather than further

optimisation on the frozen tests. A first extension is evaluation on later untouched months. A second is replication on a different mortgage or consumer-credit portfolio, especially an African dataset if the work is positioned for an African journal audience. Such a replication should redefine the outcome and macroeconomic inputs for the local product and institutional setting rather than transfer United States coefficients.

A third extension is a genuine real-time macroeconomic exercise using historical data vintages where available. A fourth is uncertainty analysis that resamples both calendar blocks and borrower clusters or uses a hierarchical design suited to repeated loan-month observations. Alternative state counts, covariance structures, duration models, or Bayesian state uncertainty may be studied prospectively with a new validation protocol. They should not be selected retrospectively from the test periods reported here.

Finally, threshold decisions, economic value, fairness, and governance require separate analysis. The present thesis evaluates probability forecasts. Approval, pricing, collections, and capital applications would need explicit costs, capacity constraints, protected-group assessment, model-risk controls, and locally relevant validation.

6.4 Closing Statement

Macroeconomic regimes offer a coherent way to organise changing credit conditions, but coherence does not guarantee stable forecast gains. In this study the regime-aware structure was transparent and occasionally advantageous, yet its benefit depended on the evaluation period and its posterior signal was nearly binary. The borrower-only model provided the stronger pooled causal forecast. The lasting result is therefore not that regimes always improve PD estimation, but that regime-aware credit models should earn their role through causal, calibration-first, temporally repeated evaluation.

Data Definitions and Transformations

A.1 Loan-Month Outcome and Risk Set

The unit of analysis is an eligible loan at calendar month t . The target $Y_{i,t}^{(12)}$ equals one when loan i first reaches at least 90 days past due in one of months $t + 1$ through $t + 12$. A row is retained only when the loan is not already at least 90 days past due at t and the complete twelve-month outcome window is observable. Rows after the first qualifying event do not re-enter the risk set. All payment-history features use information through month t only.

A.2 Loan-Level Numeric Features

Table A.1 lists the numeric variables used by the empirical borrower-only model. Names are reproduced from the implementation so that the fitted design matrix can be traced directly to the source code. Freddie Mac field meanings should be read together with the Single-Family Loan-Level Dataset documentation for the applicable release.

Table A.1: Loan-level model feature dictionary

Implementation name	Definition in this study
<code>credit_score</code>	Borrower credit score recorded at origination.
<code>mi_percent</code>	Mortgage-insurance percentage at origination.
<code>number_of_units</code>	Number of units in the financed property.
<code>original_cltv</code>	Original combined loan-to-value ratio.
<code>original_dti</code>	Original debt-to-income ratio.
<code>original_upb</code>	Original unpaid principal balance.
<code>original_ltv</code>	Original loan-to-value ratio.
<code>original_interest_rate</code>	Contract interest rate at origination.
<code>original_loan_term</code>	Original contractual term in months.
<code>number_of_borrowers</code>	Number of borrowers recorded on the loan.
<code>loan_age_months</code>	Number of months elapsed since origination.
<code>remaining_months_to_legal_maturity</code>	Contractual months remaining at the observation date.
<code>current_actual_upb</code>	Current actual unpaid principal balance.
<code>current_interest_rate</code>	Contract interest rate at the observation date.

Implementation name	Definition in this study
<code>current_delinquency_months</code>	Current days-past-due measure divided by 30 and rounded down.
<code>current_30_dpd</code>	Indicator that current delinquency is at least 30 days.
<code>current_60_dpd</code>	Indicator that current delinquency is at least 60 days.
<code>recent_30_dpd_3m</code>	Indicator for at least one 30-DPD month in the current three-month window.
<code>recent_60_dpd_3m</code>	Indicator for at least one 60-DPD month in the current three-month window.
<code>ever_30_dpd</code>	Indicator that the loan has reached 30 DPD by month t .
<code>ever_60_dpd</code>	Indicator that the loan has reached 60 DPD by month t .
<code>months_since_30_dpd</code>	Months since the latest observed 30-DPD month; before the first event, elapsed months from the loan's first panel month plus one.
<code>months_since_60_dpd</code>	Months since the latest observed 60-DPD month, defined analogously.
<code>delinquent_30_months_3m</code>	Count of 30-DPD months in the current three-month window.
<code>delinquent_30_months_6m</code>	Count of 30-DPD months in the current six-month window.
<code>delinquent_30_months_12m</code>	Count of 30-DPD months in the current twelve-month window.
<code>delinquent_60_months_3m</code>	Count of 60-DPD months in the current three-month window.
<code>delinquent_60_months_6m</code>	Count of 60-DPD months in the current six-month window.
<code>delinquent_60_months_12m</code>	Count of 60-DPD months in the current twelve-month window.
<code>cured_from_30_dpd</code>	Indicator that the previous month was at least 30 DPD and the current month is below 30 DPD.
<code>cumulative_30_dpd_cures</code>	Cumulative number of observed 30-DPD cures through month t .
<code>months_since_30_dpd_cure</code>	Months since the latest 30-DPD cure; before the first cure, elapsed months from the loan's first panel month plus one.

A.3 Macroeconomic Transformations

The empirical pipeline derives the variables in Table A.2 from monthly FRED observations. Changes and growth rates are backward-looking. Missing transformed observations created by lags are removed before HMM fitting. The regime model itself uses four adverse-oriented

factors derived from this wider set; the additive macro model retains the transformed macroeconomic vector.

Table A.2: Transformed macroeconomic feature dictionary

Implementation name	Series interpretation or transformation
<code>inflation</code>	Twelve-month percentage change in the Consumer Price Index.
<code>policy_rate</code>	Monthly federal funds rate.
<code>unemployment</code>	Monthly civilian unemployment rate.
<code>unemployment_change</code>	Month-to-month change in unemployment.
<code>house_price_growth</code>	Twelve-month percentage change in the FHFA house-price index.
<code>mortgage_rate</code>	Monthly average 30-year fixed mortgage rate.
<code>mortgage_spread</code>	Mortgage rate less the ten-year Treasury rate.
<code>long_rate</code>	Monthly ten-year Treasury rate.
<code>long_rate_change</code>	Month-to-month change in the ten-year Treasury rate.
<code>yield_curve</code>	Ten-year less two-year Treasury spread.
<code>initial_claims</code>	Initial unemployment-insurance claims, aggregated to monthly frequency.
<code>initial_claims_change</code>	Month-to-month percentage change in initial claims.
<code>payroll_growth</code>	Month-to-month percentage change in nonfarm payroll employment.
<code>real_income_growth</code>	Month-to-month percentage change in real disposable personal income.
<code>financial_conditions</code>	Chicago Fed National Financial Conditions Index.
<code>consumer_sentiment</code>	University of Michigan consumer-sentiment index.
<code>consumer_sentiment_change</code>	Month-to-month change in consumer sentiment.

Frozen Model and Validation Protocol

B.1 Core Specifications

Let $X_{i,t}$ contain the 32 variables in Table A.1, after training-window standardisation and median imputation in the fitted pipeline. The three frozen specifications were:

$$\text{logit}(\pi_{i,t}^B) = \beta_0 + X_{i,t}^\top \beta, \quad (\text{B.1})$$

$$\text{logit}(\pi_{i,t}^M) = \gamma_0 + X_{i,t}^\top \gamma + M_t^\top \delta, \quad (\text{B.2})$$

$$\text{logit}(\pi_{i,t}^R) = \alpha_0 + X_{i,t}^\top \alpha + \eta q_t + \sum_{j \in \mathcal{J}} \theta_j q_t X_{i,t,j}. \quad (\text{B.3})$$

The interaction set $\mathcal{J}$ comprised credit score, original combined LTV, original DTI, original LTV, original interest rate, current delinquency months, recent 60-DPD status over three months, and months since 30 DPD. All models were regularised logistic regressions fitted with deterministic settings. No black-box model, new threshold, or post-test feature search entered the freeze pass.

B.2 Rolling Origins

Table B.1: Frozen chronological fold boundaries

Origin	Training through	Validation	Test
2021-04	March 2020	April 2020–March 2021	April 2021–March 2022
2022-04	March 2021	April 2021–March 2022	April 2022–March 2023
2023-04	March 2022	April 2022–March 2023	April 2023–March 2024

At each origin, macroeconomic transformation and scaling parameters were fitted through the training cutoff. Five deterministic HMM initialisations (11, 23, 42, 71, and 101) were attempted, and the valid fit with highest training log likelihood was selected. Logistic regularisation was selected on training/validation data. The three final logistic models were then frozen for the test. Platt scaling was the prespecified primary calibrator and was fitted on the chronological validation block. Isotonic calibration was not selected using rolling-test results.

B.3 Selected Hyperparameters and Calibration Maps

For raw probability p , the Platt map was

$$\hat{p} = \text{logit}^{-1}\{a + b \text{logit}(p)\}, \quad (\text{B.4})$$

where a and b in Table B.2 were estimated only on the corresponding validation block.

Table B.2: Selected regularisation and validation-fitted Platt parameters

Origin	Model	α	Platt intercept a	Platt slope b
2021-04	Borrower-only	3×10^{-5}	-0.833296	0.852419
	Macro-augmented	1×10^{-5}	-0.019378	0.271140
	Regime-aware	3×10^{-5}	-1.494595	0.734275
2022-04	Borrower-only	3×10^{-5}	-1.731286	0.863677
	Macro-augmented	1×10^{-5}	1.607486	0.923672
	Regime-aware	1×10^{-5}	-1.081113	1.005986
2023-04	Borrower-only	1×10^{-5}	-1.389040	0.946095
	Macro-augmented	1×10^{-5}	-1.413346	0.768916
	Regime-aware	1×10^{-5}	-1.490654	0.887863

Additional Rolling-Origin Diagnostics

C.1 Validation Performance

Table C.1 reports the chronological validation metrics used before each test. Regime-aware had the lowest validation Brier score and log loss at each origin, while borrower-only was better on some validation ranking measures. The later test reversals therefore cannot be attributed to selecting the regime-aware model from test outcomes; they show that validation ordering did not always transport.

Table C.1: Validation performance in the causal rolling pass

Origin	Model	Brier	Log loss	ROC-AUC	PR-AUC
2021-04	Borrower-only	0.017515	0.080816	0.822049	0.344965
	Macro-augmented	0.022265	0.108478	0.515855	0.029321
	Regime-aware	0.017156	0.078880	0.824940	0.374962
2022-04	Borrower-only	0.004666	0.026231	0.793017	0.138926
	Macro-augmented	0.004730	0.026953	0.776165	0.119958
	Regime-aware	0.004624	0.025584	0.824731	0.153336
2023-04	Borrower-only	0.004774	0.026677	0.805408	0.111178
	Macro-augmented	0.004814	0.027625	0.767303	0.104784
	Regime-aware	0.004734	0.026091	0.826694	0.122011

C.2 Pooled Calibration Deciles

Each row in Tables C.2 and C.3 contains approximately one tenth of the pooled rolling-test loan-months. Deciles were formed from each model’s own pooled predictions. The upper decile contains most observed events and also the largest absolute overprediction.

The borrower-only calibration curve was close to the diagonal in its first five deciles and increasingly high thereafter. The regime-aware model underpredicted in the lowest four deciles and overpredicted from the sixth decile upward. These grouped values explain why a useful ranking signal can coexist with poor calibration-in-the-large.

C.3 HMM Initialisation Audit

The selected seeds were 101, 71, and 71 at the three origins. At origin 2021-04, starts 23 and 71 were invalid because one transition-matrix row had no outgoing transition and therefore

Table C.2: Borrower-only pooled rolling-test calibration by prediction decile

Decile	Observations	Mean PD	Event rate
1	665,086	0.000661	0.000630
2	665,086	0.001147	0.001046
3	665,086	0.001629	0.001693
4	665,086	0.002234	0.002197
5	665,086	0.003073	0.003031
6	665,086	0.004344	0.003609
7	665,086	0.006448	0.004454
8	665,086	0.010313	0.005183
9	665,085	0.018273	0.007316
10	665,085	0.061912	0.026222

Table C.3: Regime-aware pooled rolling-test calibration by prediction decile

Decile	Observations	Mean PD	Event rate
1	665,086	0.000504	0.000908
2	665,086	0.001046	0.001352
3	665,086	0.001614	0.001973
4	665,086	0.002388	0.002854
5	665,086	0.003619	0.003786
6	665,086	0.005678	0.004279
7	665,086	0.008907	0.004342
8	665,086	0.013784	0.005085
9	665,085	0.022324	0.006981
10	665,085	0.064058	0.023821

did not sum to one. They were excluded before selection. The other three starts at that origin and all five starts at the later origins produced valid transition matrices. Selection used training log likelihood only; validation and test defaults did not enter HMM selection.

The extreme stress shares in the three test windows were 1.0000, 1.0000, and 0.9167. The corresponding hard-stress shares were 0.0833, 1.0000, and 0.6667. Mean binary entropy was effectively zero in the first two tests and 0.0376 in the third. These diagnostics support use of the state as a sharply classified period indicator while limiting continuous-probability and hard-regime comparisons.

C.4 Paired Temporal Uncertainty

Monthly borrower-only and regime-aware losses were paired before resampling. Ten thousand deterministic moving-block replicates sampled blocks of three consecutive months. The regime-aware-minus-borrower-only Brier estimate was 0.0000869 with a 95% interval of $[-0.0001558, 0.0002714]$; 25.18% of replicates favoured regime-aware. The log-loss estimate was 0.0018573 with interval $[-0.0008756, 0.0040205]$; 8.76% favoured regime-aware. These are uncertainty summaries for temporal test blocks, not confidence intervals for a universal population parameter.

Reproducibility and Artefact Manifest

D.1 Execution Boundary

The freeze pass is implemented in `src/rs_hmm/rolling_freeze.py` and invoked by `scripts/run_rolling_freeze.py`. The data-bearing execution used the repository's existing empirical inputs and fixed model definitions. No test outcome was used to change model families, interaction lists, thresholds, HMM state count, covariance form, or calibration method.

The worktree used to prepare this document does not contain the large ignored raw and processed datasets. The verified empirical run was therefore executed in the data-bearing primary checkout at `/Users/theboyyy/Work/Research/RS-HMM-Credit-Risk`. This is an environment-location limitation, not a difference in tracked code. The resulting tables were regenerated twice, and all eleven CSV artefacts were byte-identical across the two runs.

D.2 Commands

From a configured data-bearing checkout, the frozen run is invoked with:

```
python scripts/run_rolling_freeze.py
```

The focused and full automated checks are:

```
pytest -q tests/test_rolling_freeze.py  
pytest -q
```

The verified full suite reported 27 passing tests. The deterministic SGD fits emitted fixed-iteration convergence warnings at the configured 30 iterations; they completed and reproduced byte-identical result artefacts. Those warnings should be retained in the execution record and not misreported as failed fits.

The thesis PDF is compiled from the `manuscript` directory using the approved isolated TeX Live procedure documented with the project. Bibliographic processing is included in that compilation sequence.

D.3 Authoritative Rolling Artefacts

Table D.1: Frozen rolling-origin result artefacts

CSV file under reports/tables	Contents
<code>rolling_origin_folds.csv</code>	Training, validation, and test boundaries.
<code>rolling_origin_hmm_initialisations.csv</code>	Candidate seeds, training likelihoods, convergence status, and errors.
<code>rolling_origin_hmm_paths.csv</code>	Origin-specific filtered stress paths and selected seed.
<code>rolling_origin_stress_diagnostics.csv</code>	Posterior range, hard/extreme shares, and entropy.
<code>rolling_origin_model_selection.csv</code>	Selected regularisation and Platt parameters.
<code>rolling_origin_coefficients.csv</code>	Standardised fitted logistic coefficients.
<code>rolling_origin_metrics.csv</code>	Fold-level validation and test metrics.
<code>rolling_origin_calibration_bins.csv</code>	Fold and pooled grouped calibration.
<code>rolling_origin_monthly_metrics.csv</code>	Monthly losses, event rates, and predicted-default counts.
<code>rolling_origin_paired_monthly_losses.csv</code>	Monthly regime-aware-minus-borrower-only paired losses.
<code>rolling_origin_paired_uncertainty.csv</code>	Moving-block estimates, intervals, and replicate settings.

D.4 Reproduction Checklist

An independent reproduction should verify the input file inventory and hashes; confirm the 2015–2024 cohort selection; rebuild payment-history variables in loan and month order; check that every outcome has complete forward follow-up; inspect chronological split boundaries; confirm that scalars and HMM parameters are training-only at each origin; compare selected HMM seeds and likelihoods; fit the three frozen logistic models; fit Platt calibration only on validation; and compare all eleven CSV artefacts. A substantive replication should additionally document software versions, processor architecture, and any numerical-library differences.

D.5 Items Requiring Author or Supervisor Confirmation

Before institutional submission, the author and supervisors must confirm the official title-page wording, submission month and year, declaration and certification format, dedication, acknowledgements, degree and department naming, and any required ethical-review or data-use statement. They should also approve whether an unreviewed 2026 contextual preprint may be cited; it is deliberately absent from this thesis bibliography. For a later journal manuscript, the target journal's current scope, formatting rules, data-availability wording, authorship order, affiliations, funding disclosure, conflict-of-interest statement, and use of United States data for an African readership must be reviewed separately.

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